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GABES

Graphical & Accelerated Boundary Element Solver

User's Guide

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1 INTRODUCTION

GABES (**G**raphical & **A**ccelerated **B**oundary **E**lement **S**olver) is an open-source application based on the Boundary Element Method (BEM), accelerated using an $\mathscr{H}$-Matrix library, for solving three-dimensional linear elastostatic problems in isotropic, homogeneous media. The application runs exclusively on 64bit Windows platforms and provides a user-friendly graphical interface with an interactive 3D environment. Users can import meshed 3D models, define numerical and physical parameters, apply boundary conditions interactively, and launch simulations. GABES also includes post-processing tools for graphical visualization of results and the generation of tabulated reports.

2 DOWNLOAD AND SETUP

GABES is available as an open-source package on the GitHub platform, with the repository accessible at https://github.com/rahimsihadjmohand-png/GABES. The repository contains the latest committed source code provided as a Visual Studio solution/project, as well as several binary releases. Detailed instructions for setting up the source code and installing the binary releases are presented in the following sections.

2.1 Download the latest GABES release

The GABES project is currently under active development. To facilitate immediate testing and practical use, each incremental update is published as a direct release. This approach allows users to download and run the application without compiling the source code manually.

How to Download a Release

1. Navigate to the Releases section on the GABES main repository page (see Figure 1).
2. A list of all available releases will be displayed (see Figure 2).
3. Click on the latest release (for example, Version 1.0.0) to open its dedicated page. The most recent release can also be accessed directly from the main repository page, as illustrated in Figure 1.
4. Download the release package by selecting the first .zip file (for example, GABES.v1.0.0.zip) as shown in Figure 3. This archive contains:
   - GABES.exe (the main executable file)
   - All required dependency **(*.dll)** files necessary for proper execution

Important note about source code

The release page also provides the complete source code corresponding to that specific version. However, note that the source code may sometimes be updated in the repository without publishing a new official release. Therefore, to ensure obtaining the most recent version of the source code, it

is recommended to download it directly from the repository's main page. This process will be explained in the following sections.

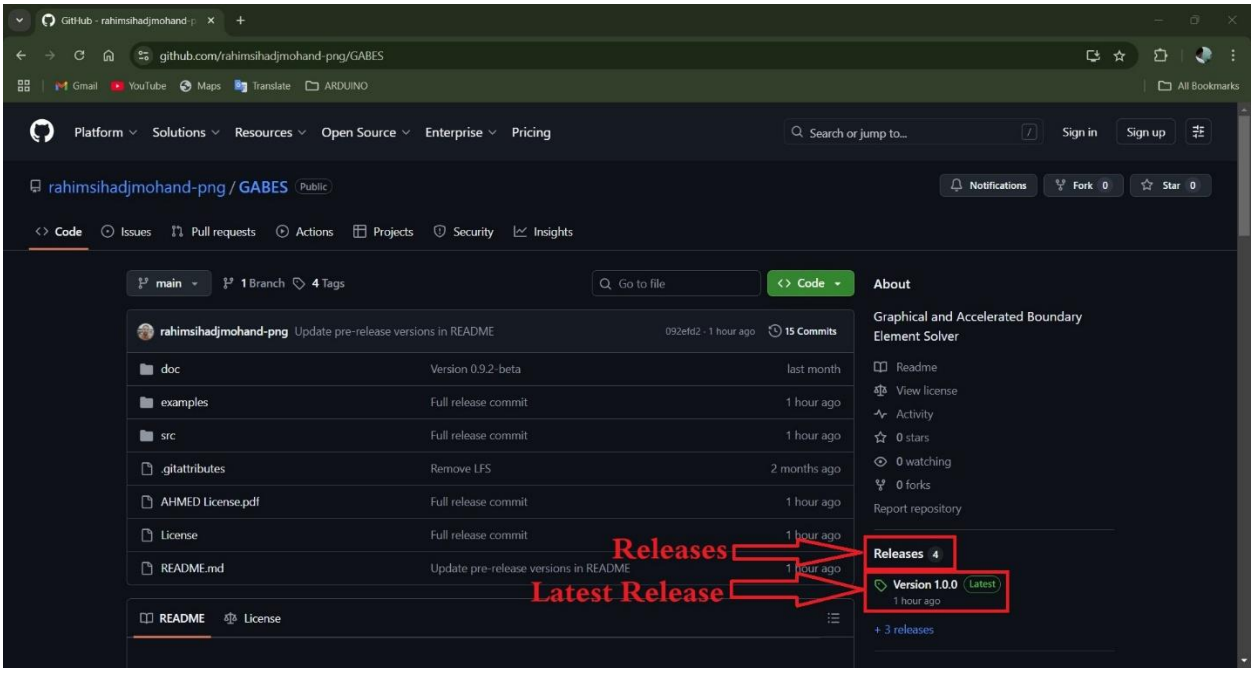


Figure 1: Accessing the releases

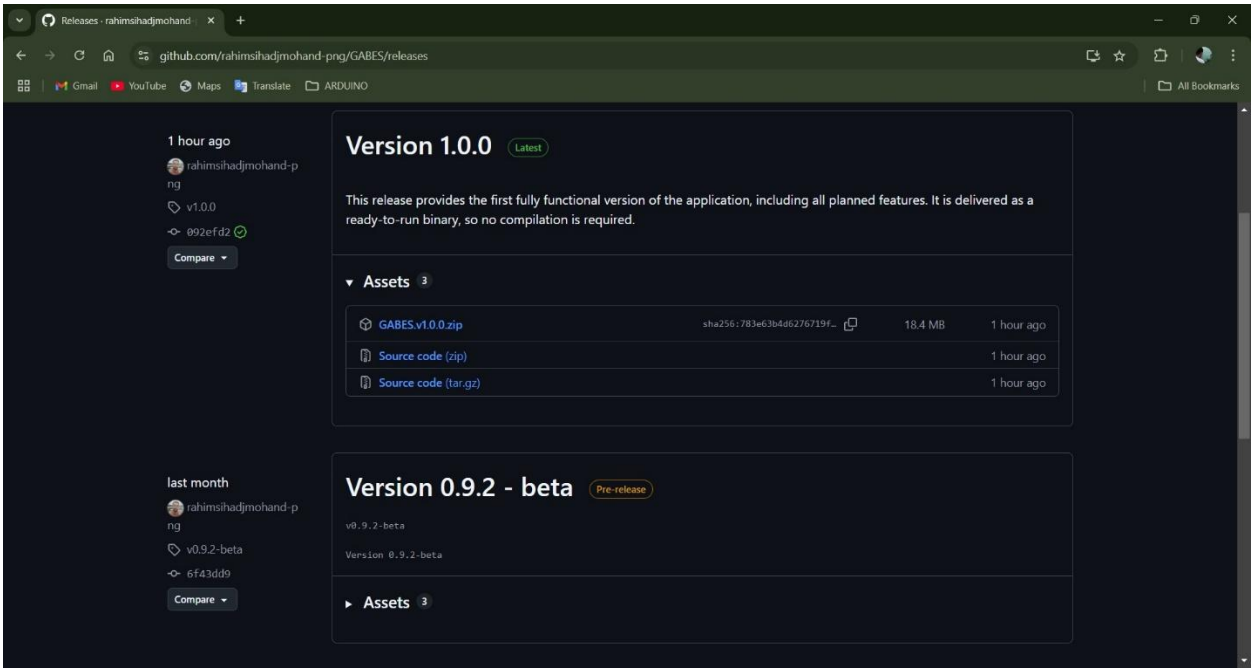


Figure 2: The available releases

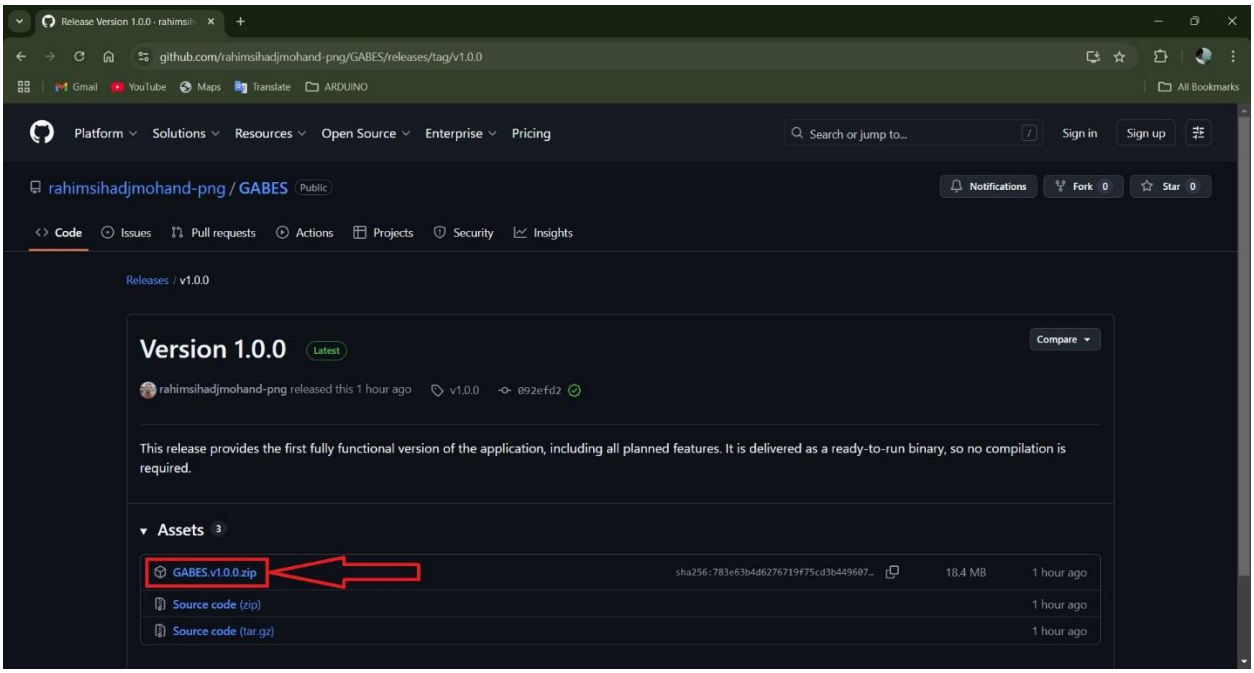


Figure 3: Latest release download

2.2 Download the source code

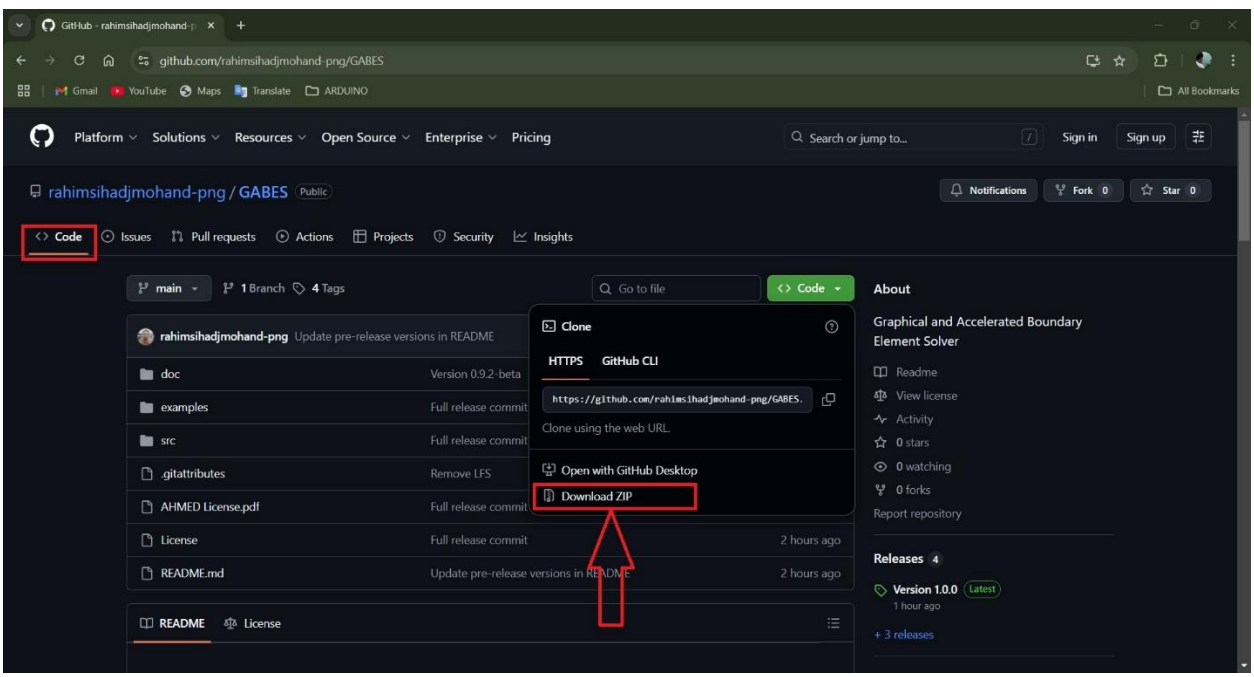


Figure 4: Download the source code

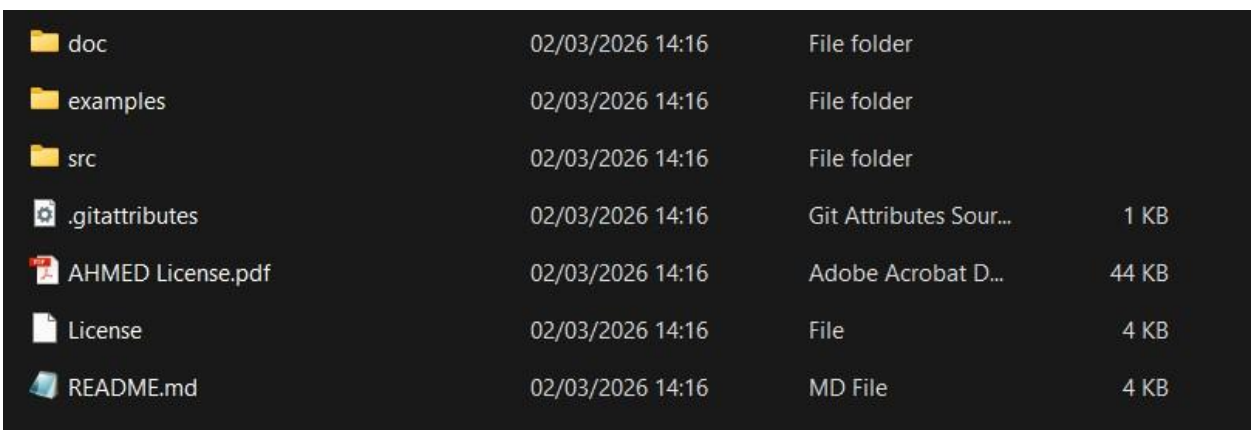
As shown in Figure 4, to download the full source code package, navigate to the **Code** tab. Click the green **Code** button; a context menu will appear. Select **Download ZIP** to start the download automatically. The ZIP archive will be saved to the browser's default download directory.

Alternatively, users with Git installed on Windows can download the repository via the command line, by opening a command prompt or terminal in the desired directory and execute the command:
`git clone https://github.com/rahimsihadjmohand-png/GABES.git`

However, for most Windows users, the first method is sufficient and more practical.

Upon completion of the download and extraction of the archive, the directory structure is organized as illustrated in Figure 5 and consists of the following contents:

1. The **doc** directory contains the present user's manual.
2. The **examples** directory contains sample models ready for execution, along with their related files.
3. The **src** directory contains the complete source code of GABES provided as a Microsoft Visual Studio solution.
4. The **.gitattributes** file contains GitHub configuration instructions.
5. The **AHMED License.pdf** file is a verbatim copy of the AHMED $\mathscr{H}$-Matrix library license.
6. The **License** file specifies the main license terms governing GABES.
7. The **README.md** file provides brief setup and usage instructions.



doc	02/03/2026 14:16	File folder	
examples	02/03/2026 14:16	File folder	
src	02/03/2026 14:16	File folder	
.gitattributes	02/03/2026 14:16	Git Attributes Sour...	1 KB
AHMED License.pdf	02/03/2026 14:16	Adobe Acrobat D...	44 KB
License	02/03/2026 14:16	File	4 KB
README.md	02/03/2026 14:16	MD File	4 KB

Figure 5: Downloaded source package

2.3 Source code modification and compilation

To successfully modify and compile the source code, the installation of the **Microsoft Visual Studio IDE** with C++ and MFC support is required. The source code is provided as a Visual Studio solution/project package that relies on several external dependencies, which must be pre-installed and properly configured within the project properties.

To open the (solution/project) package in visual studio, click on the **GABES.sln** file inside the **src** folder which opens the entire code in Visual studio as shown in Figure 6.

The configuration process is end-user dependent due to possible variations in third-party library installation paths. This configuration enables Visual Studio to correctly locate the binary dependencies as well as the directories containing the required library header files.

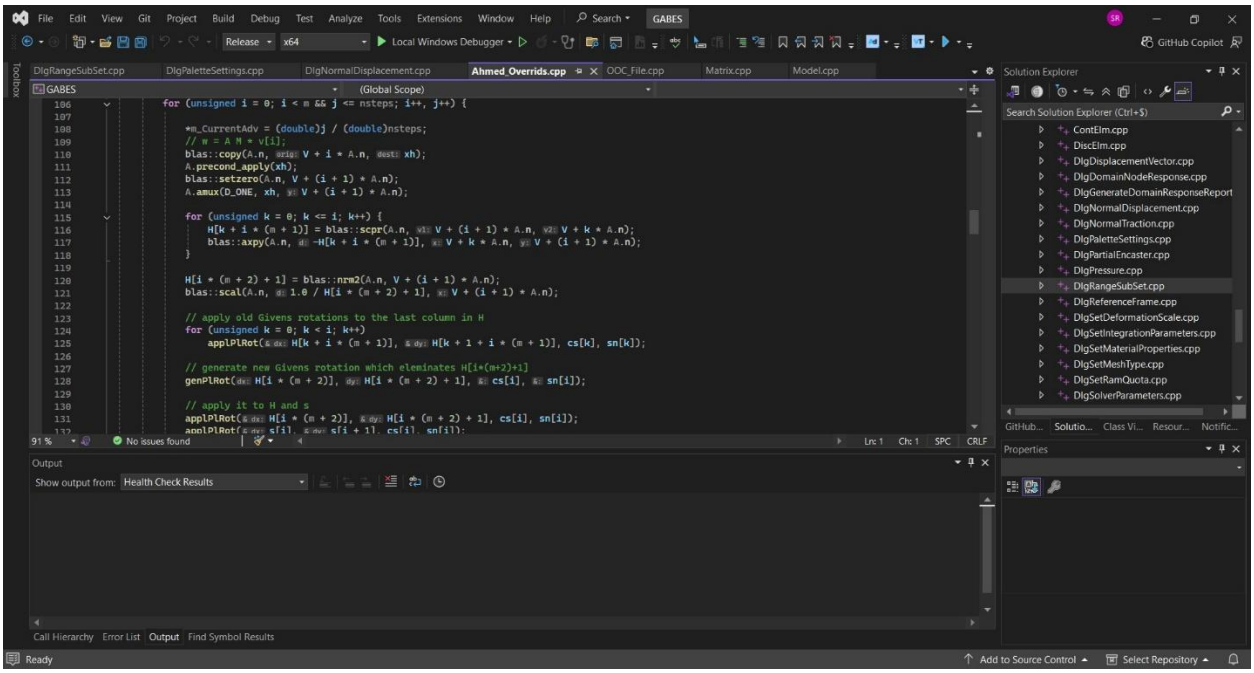


Figure 6: The GABES Visual Studio solution/project

2.3.1 Third-party dependencies

GABES integrates several third-party libraries that support different components of the application, including the graphical user interface, the 3D visualization environment, and the $\mathscr{H}$-matrix framework. Some of these libraries are natively available in standard Windows installations, while others are automatically provided or can be integrated directly through Visual Studio. In contrast, certain libraries must be installed separately, requiring manual configuration of the **include** and **library** paths by the user. All libraries used in GABES are listed in Table 1.

Table 1: GABES third-party dependencies

Library	Description	Configuration
MFC	**M**icrosoft **F**oundation **C**lasses (MFC): a C++ wrapper library for Windows graphical user interface (GUI) and desktop application development.	The library is an integral part of the Microsoft Visual Studio toolchain and is automatically configured. In older Visual Studio distributions, it is installed by default. However, in recent Visual Studio versions, MFC installation must be explicitly selected during the Visual Studio setup process; otherwise, it can be added later using the Visual Studio Installer.
DirectX 9.0	a Microsoft-specific 3D multimedia library used for 3D programming in	On older Windows distributions, DirectX must be manually installed by the user. On recent Windows

	multimedia applications, particularly games.	versions, the core DirectX runtime components are included with the operating system; however, certain legacy DirectX components (such as D3DX libraries) may still require separate installation.
AHMED	The AHMED library: provides the $\mathcal{H}$-Matrix framework, including routines for block partitioning, construction of $\mathcal{H}$-Matrix approximations, and various $\mathcal{H}$-Matrix linear algebra and solver operations.	The library must be installed, and both its header files and compiled binaries must be properly configured for the source code.
OpenMP	OpenMP is a parallel programming framework used for thread-level parallelism on multi-core, shared-memory machines. It allows programs to execute multiple threads concurrently within a single process. OpenMP does **not** provide parallelism across a network of separate machines (cluster-level parallelism).	OpenMP is available in the Visual Studio toolchain. However, its support must be enabled in the Visual Studio project settings.
BLAS	Basic Linear Algebra Subprograms (BLAS) is a standard library providing highly optimized routines for vector and matrix operations, including dot products, vector scaling, and matrix multiplication. It is widely used as a foundational library in scientific computing and high-performance numerical applications.	The BLAS library must be installed separately, and the project must be configured to include its header files and link to the compiled binaries

2.3.1.1 DirectX 9.0

GABES relies on the DirectX 9.0 Software Development Kit (June 2010), which provides all the development tools required for 3D multimedia applications. This includes the core runtime, helper libraries such as D3DX, complete header files for function prototypes, and tools for tasks such as texture management. The SDK also contains numerous tutorial examples to facilitate learning and development. The DirectX SDK can be obtained from Microsoft's official website: https://www.microsoft.com/en-us/download/details.aspx?id=6812. Installation is straightforward and guided through a graphical Windows installer, which is familiar to most users.

2.3.1.2 AHMED

GABES uses the AHMED library primarily to transform dense linear system matrices into $\mathcal{H}$-Matrices and to provide specialized $\mathcal{H}$-Matrix solver methods. These solvers have reduced algorithmic complexity, making relatively large-scale simulations feasible on a personal computer. Unlike classical direct solvers, which have cubic asymptotic complexity, the $\mathcal{H}$-Matrix technique can achieve near-logarithmic complexity, resulting in a significant computational gain.

AHMED is an open-source library distributed under an academic-only license and is accessible at the following GitHub repository: https://github.com/xantares/ahmed.git. It is provided as a cross-platform CMake C++ project. To set it up on Windows, the AHMED CMake project can be configured within Visual Studio with Windows-specific parameters to generate a Visual Studio solution package. Compiling this solution produces the final **.dll** and **.lib** files. The **.lib** files are specifically required in Microsoft Visual C++ to correctly link and access the corresponding dynamic-link library **(.dll)**, which contains the main library code. For Windows, the provided AHMED **CMakeLists.txt** file should be carefully reviewed and adapted to ensure successful compilation. Once the Visual Studio AHMED project is built successfully, it will generate the required files (**AHMED.dll** and **AHMED.lib**), which are then used in the GABES project.

2.3.1.3 BLAS

BLAS (Basic Linear Algebra Subprograms) is a highly optimized library used for fundamental vector and matrix operations, including dot products, vector scaling, and matrix multiplication. It is widely used in scientific computing and high-performance numerical applications to accelerate core numerical routines. Common BLAS implementations include **OpenBLAS** and **Intel Math Kernel Library (MKL)**. OpenBLAS is an open-source, high-performance implementation available at https://github.com/OpenMathLib/OpenBLAS, and Intel MKL is a commercially supported, highly optimized implementation available as part of **Intel oneAPI**.

GABES uses the Intel MKL implementation of BLAS in the AHMED library's overridden functions. But The user is free to use another implementation of BLAS.

2.3.2 Visual Studio project configuration

2.3.2.1 Configuration of the library paths

The Microsoft Visual C++ compiler (MSVC) requires explicit specification of the locations of all library header and binary files. These paths are configured through the project's **Properties** dialog, which can be accessed from the Visual Studio main menu by selecting **Debug**, then **GABES Debug Properties**.

Within the **Configuration Properties** tree, the user must navigate to **VC++ Directories**. For each external library, both the *Include* and *Library* directories must be specified by selecting **Edit** from the corresponding dropdown combo-box. Consequently, the user must have prior knowledge of the locations of the library header files, which are added to **Include Directories**, as well as the binary (.lib) files, which are added to **Library Directories**.

In most cases, header files are located in an **\include** directory, or occasionally in a **\src** directory for projects that do not separate source (.c, .cpp) files from header (.h) files. Binary library files are generally located in a **\lib** directory; however, care must be taken to select the correct platform-specific subdirectory. Typically, libraries provide separate folders for 32-bit (x86) and 64-bit (x64) builds. It is important to note that **GABES targets exclusively 64-bit platforms**, and no configuration is provided for 32-bit builds.

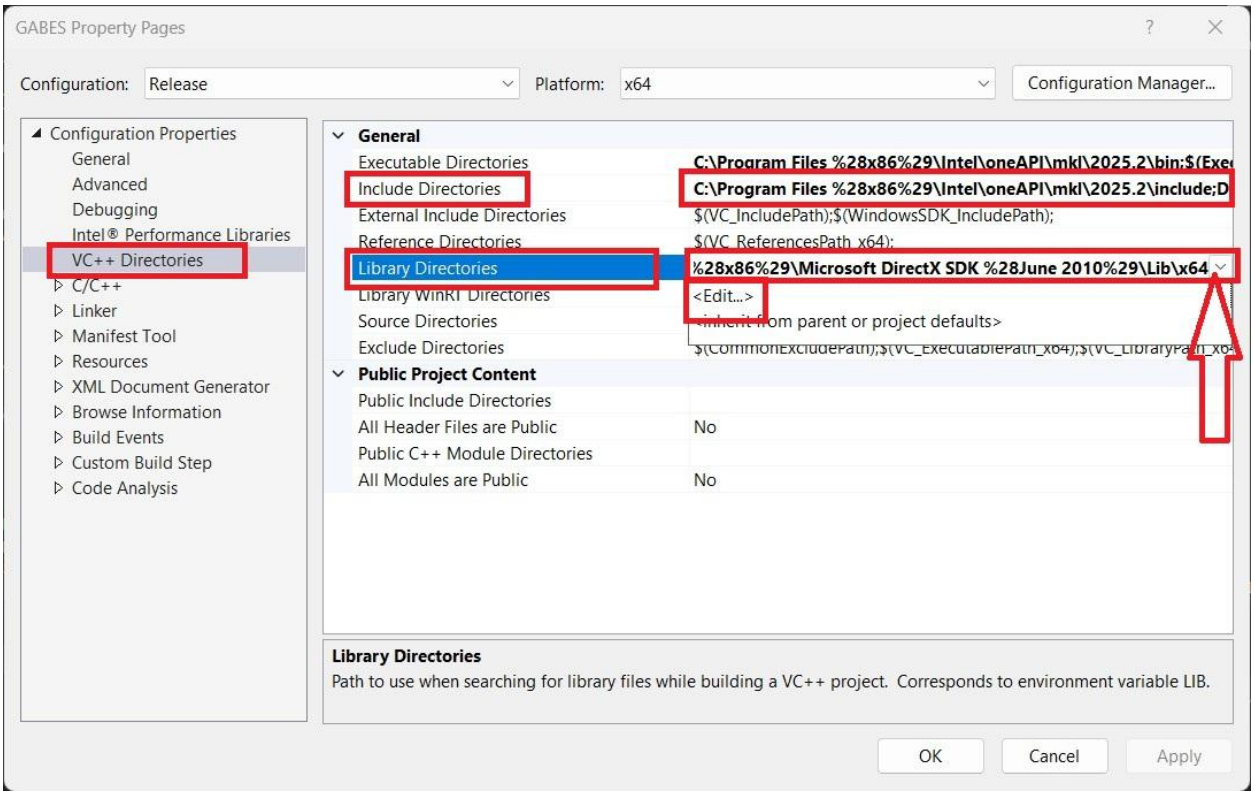


Figure 7: Library paths configuration

2.3.2.2 Specifying linker input library files

Once the library paths have been configured, the next step is to explicitly tell the MSVC linker which **(.lib)** import-library files it should use when building the final executable **(.exe)**. This configuration is performed within the same **Project Properties** dialog.

As shown in Figure 8, navigate to: **Linker** > **Input**. In the **Additional Dependencies** field, you must add the required import-library files. This can be done by clicking the corresponding dropdown combo-box and entering the import-library name. In the dialog that appears, add the library files listed in Table 2.

Table 2: Required linker input library files

Import-library Files	**Target Library**
d3d9.lib	Direct3D 9 Core Library
d3dx9.lib	Direct3D 9 Extensions Library (D3DX)
ahmed.lib	AHMED $\mathscr{H}$-Matrix library
mkl_core.lib	Intel Math Kernel Library (MKL) Core
mkl_intel_lp64.lib	MKL LP64 Interface (for 64-bit integers)
mkl_sequential.lib	MKL Sequential Layer
mkl_intel_thread.lib	MKL Intel Threading Layer

N.B. If the BLAS functionality is supplied by a library other than Intel MKL, the required linker files will differ accordingly. For instance, when using OpenBLAS, you would substitute all MKL libraries with **libopenblas.lib**.

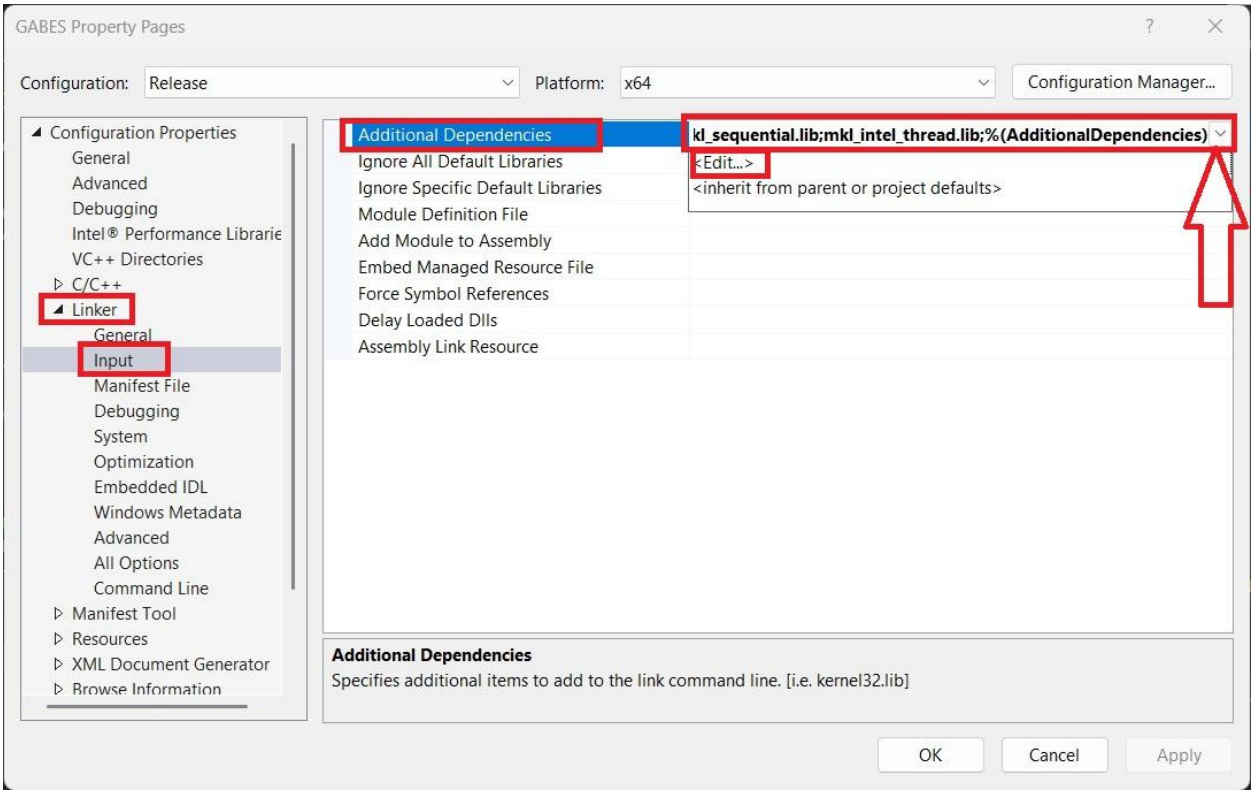


Figure 8: Specifying linker input library files

2.3.2.3 Activating OpenMP

OpenMP is used for thread-level parallelism (machine-local parallelism) and is primarily employed in GABES to parallelize loops involved in the computation of the coefficient matrix. OpenMP is already integrated into Visual Studio, but it must be explicitly enabled in the project properties.

To activate OpenMP, go to the Visual Studio main menu, select **Debug**, then choose **GABES Debug Properties**. In the project properties window, expand the **C/C++** tree, select **Language**, and set the **OpenMP Support** field to **Yes (/openmp)**, as illustrated in Figure 9. Users should ensure that the correct build configuration (Debug or Release) is selected, since each configuration can have different compilation options.

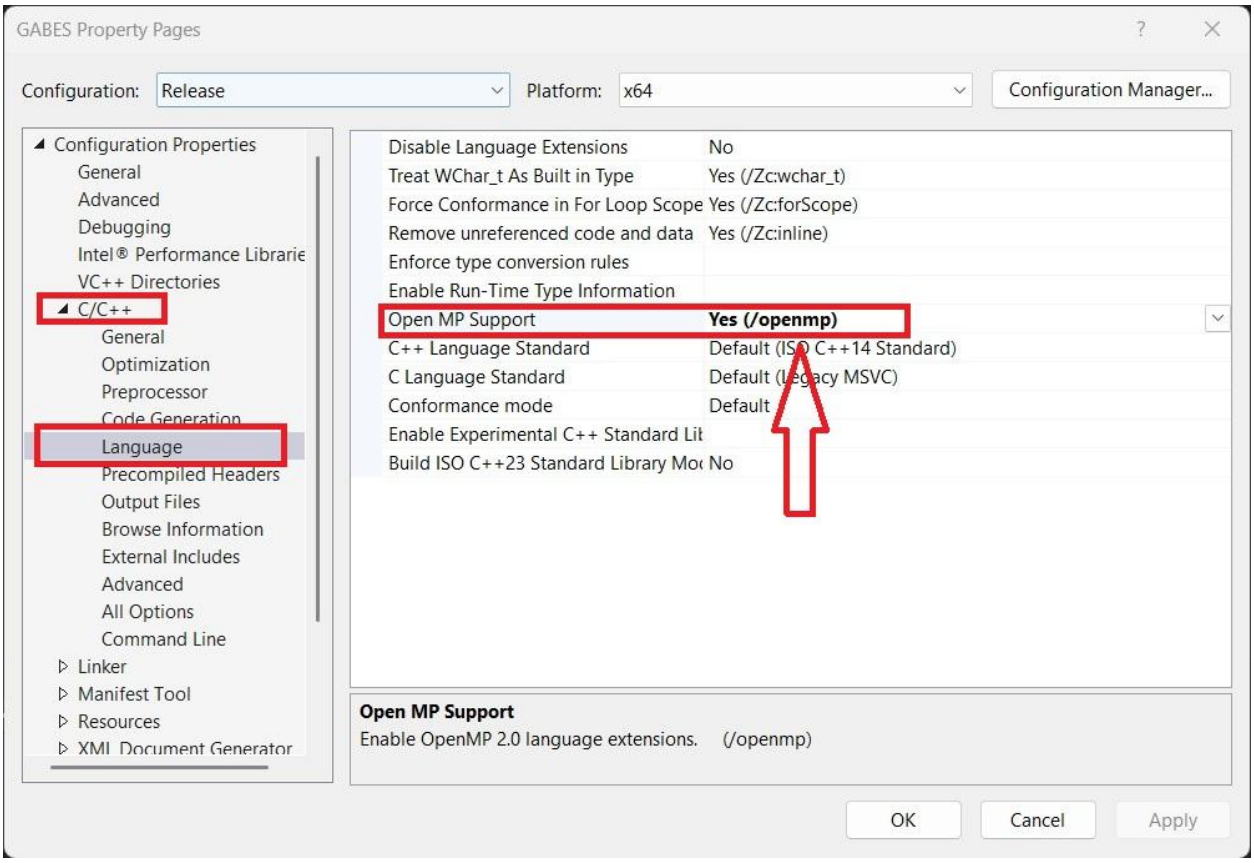


Figure 9: OpenMP Enabling

3 PRESENTATION OF THE APPLICATION

3.1 The Mainframe Window

The **Mainframe** window serves as the container for all graphical components of the application, as illustrated in Figure 10. It includes:

- **Client Area:** the primary workspace for 3D visualization and interaction;
- **Ribbon**: the main toolbar for commands and tools;
- **View Toolbar**: a toolbar for commands and tools to manipulate the 3D model;
- **Model Tree**: a hierarchical display of the various objects used during the modeling phase;
- **Information Panel**: displays context-specific data and parameters;
- **Status Bar**: shows the coordinates and the indices of the selected elements and nodes.

Each of these components will be described in more detail in the following sections.

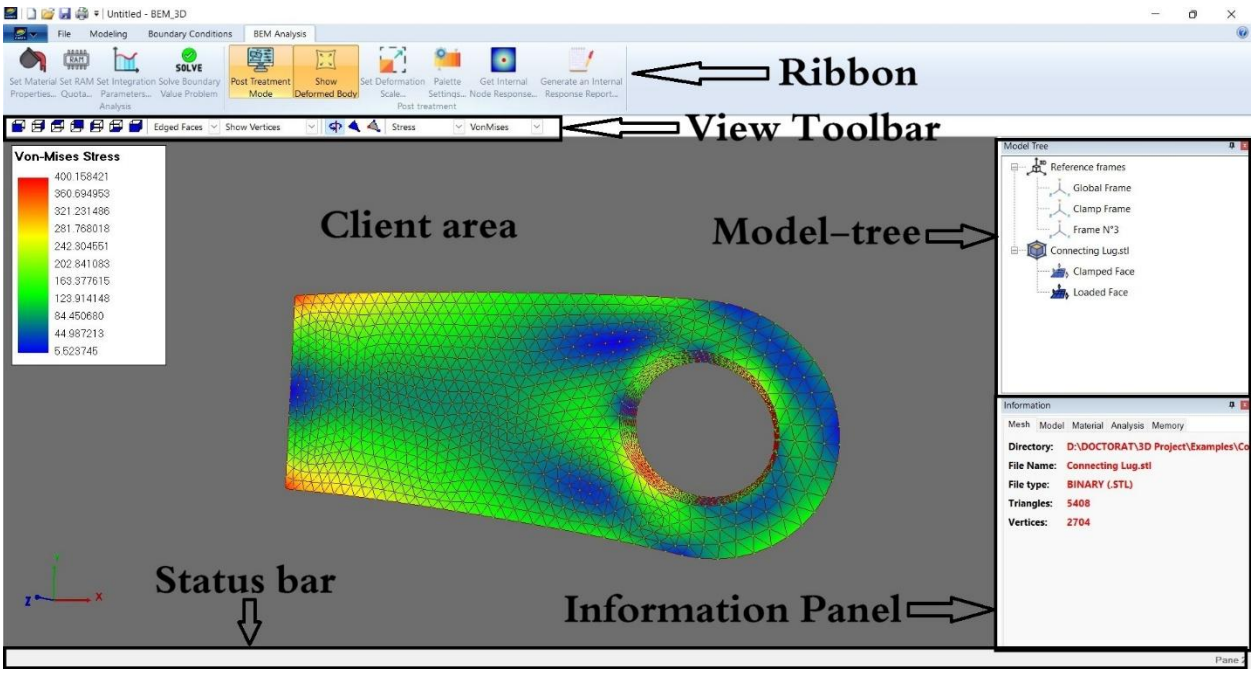


Figure 10: Mainframe Window

3.2 The Client Area

The **Client Area** is the primary space for visualizing the 3D model in both the modelling and post-processing phases. It permits rotation and zooming of the 3D model and contains a small representation of the three coordinate axes (Oxyz), known as the **Gizmo**.

In the **modelling phase**, the Client Area allows interactive selection of triangular mesh elements, vertices, and DOF nodes, which can be useful for various purposes. It also displays several objects (see Figure 11):

- The imported model mesh
- The created coordinate frames
- The created surface subsets
- The applied boundary conditions (simple/double fixations, displacement vectors, and traction vectors)

In the **post-processing phase** (see Figure 12), it displays:

- Various color-mapped field variables (displacements, stresses, and strains)
- A color palette with the corresponding values
- The deformed and undeformed configuration of the body

The model can be visualized in different output modes, including **solid surface**, **edged surface**, and **wireframe**. The visibility of geometrical mesh vertices and DOF nodes can be toggled on or off. All these view options can be controlled and configured using the **View Toolbar**.

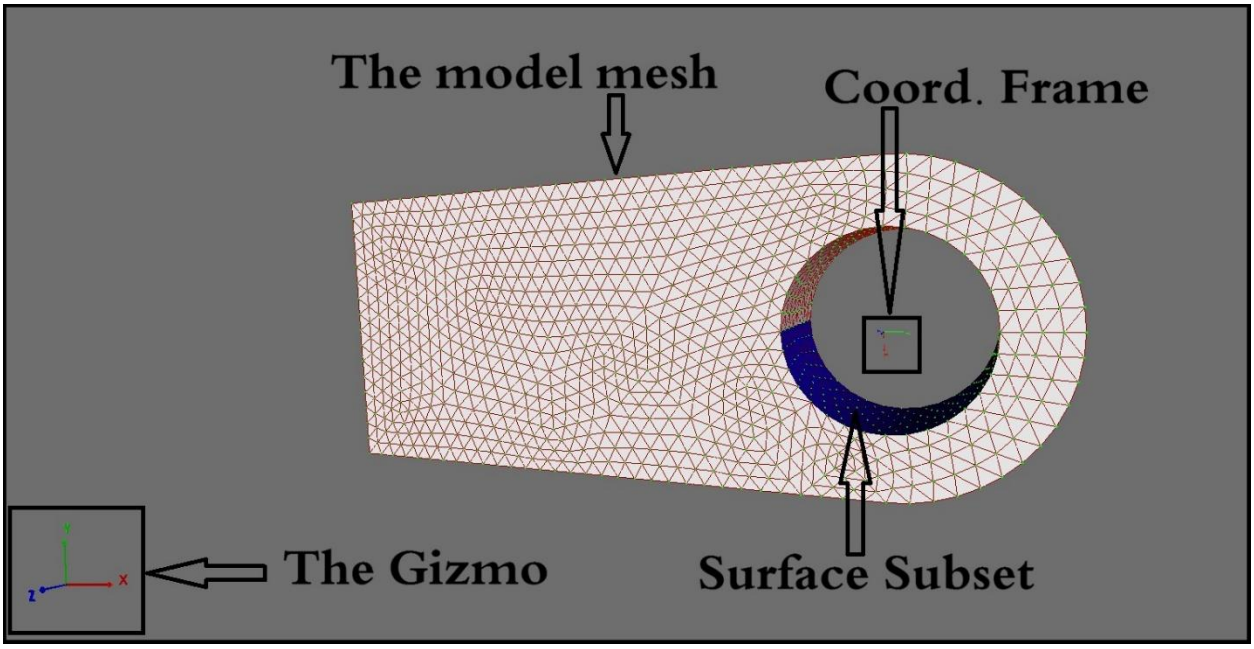


Figure 11: Client Area modeling phase

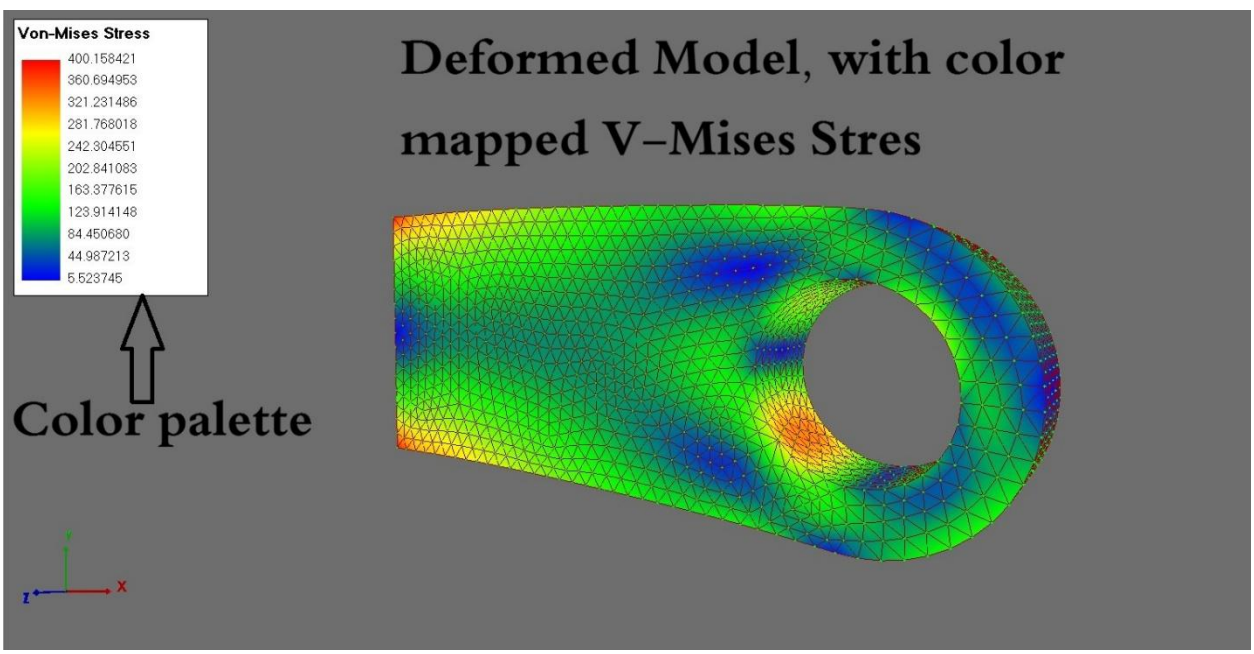


Figure 12: Client Area post-processing phase

3.3 The Ribbon

Modern GUI applications feature a **Ribbon**, which replaces the traditional menu and toolbar system. Commands are provided through buttons with larger icons, organized into panels across multiple separate tabs, making them more flexible and easily accessible.

The majority of **GABES's** commands are embedded in the **Ribbon**, which is organized into four (04) tabs:

3.3.1 File tab

Subdivided into two (02) panels:

- **Geometry panel:** Contains two (02) commands for loading and deleting the model mesh
- **GABES File Management panel:** Contains four (04) commands for creating, opening and saving GABES files.

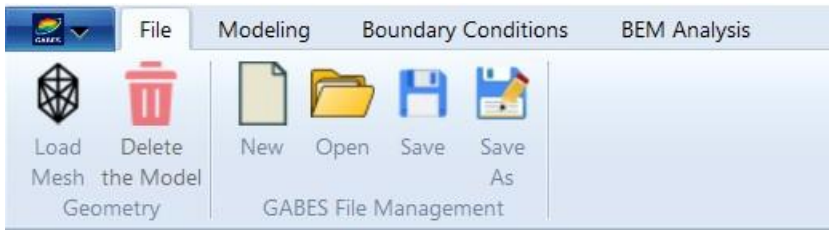


Figure 13: File tab

The **File tab** commands are summarized in Table 3

Table 3: Command buttons of the File tab

Panel	Button	Action
Geometry	Load Mesh	Opens a standard Windows **Open File** dialog to navigate to the location of the mesh file, which can be in one of two formats: **.stl** or **.obj**. The 3D surface mesh is then loaded into **GABES**
	Delete the Model	Deletes the current model and cleans up all generated data
GABES File Management	New	Creates a new **GABES** file and may prompt the user to save the current model if modifications have occurred
	Open	Opens an existing **GABES** file **(*.gbs)**, which may contain all modeling information, including the mesh, reference frames, surface subsets, boundary conditions, and field variable results.

	Save	Saves the current model to an existing **GABES** file, or prompts the user to choose a save location if the file has not yet been created
	Save As	Saves the current model to a new disk location by prompting the user to choose the destination directory.

3.3.2 Modeling tab

The tab is subdivided into two (02) panels:

- **Frames panel:** Contains a command for creating new reference frames, which are very helpful during surface subset creation and post-processing report generation.
- **Surface Subsets panel:** Contains three (03) different commands for generating surface subsets, which are mandatory for boundary condition assignment.

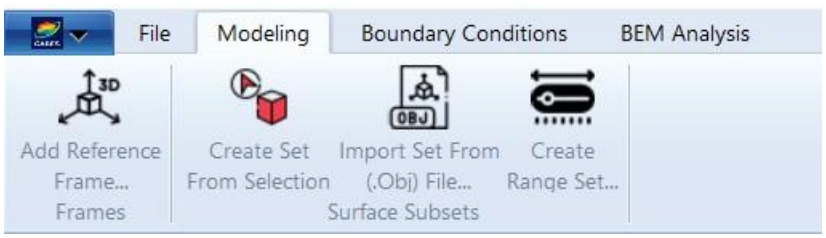
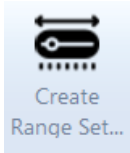


Figure 14: Modeling tab

The **Modeling tab** commands are summarized in Table 4

Table 4: Command buttons of the Modeling tab

Panel	Button	Action
Frames	Add Reference Frame...	Opens a modeless dialog box that allows the graphical, interactive creation of a new frame by defining the location of the origin point and the orientation of the axes.
Surface Subsets	Create Set From Selection	Creates a new surface subset from the currently selected elements. This button becomes active only when a valid selection exists **N.B.** *Elements can be selected interactively in the 3D model using the mouse cursor, with the **CTRL** key enabling multiple selections.*
	Import Set From (.Obj) File...	This command opens a standard Windows **Open File** dialog to navigate to the location of a **.obj** file describing the subset geometry. This method can be used with meshes created in **Autodesk 3ds Max**, where individual surfaces can be selected and detached using the 3ds Max **Detach**

		command. These surfaces can then be exported as Wavefront **.obj** files, which **GABES** can read and convert into surface subsets. **N.B.** *This method is widely adopted in the **BESLE** software.*
		This command implements the third method in GABES for creating surface subsets. It uses previously defined coordinate frames to map model coordinates into the selected frame within one of three coordinate systems: Cartesian, cylindrical, or spherical. The user can then specify coordinate ranges either by entering intervals manually or by using the provided slider bars, with selections updating graphically in real time. This method is particularly efficient for well-structured geometries with planar, cylindrical, or spherical sections, which is typical for most mechanical parts.

3.3.3 Boundary Conditions tab

The tab is subdivided into two (02) panels:

- **Displacement BCs:** Contains commands for assigning various Dirichlet-type boundary conditions, including total clamping, partial clamping (zero displacement in selected directions), vectorial displacement, and normal displacement. This panel also provides check boxes to toggle the visibility of the created fixations and displacement vectors.
- **Traction BCs:** Contains commands for assigning various Neumann-type boundary conditions, including vectorial traction, normal traction, and pressure. It also provides the **Free Surface** command to release a surface from the previously assigned boundary conditions and restore it to a traction-free state. A checkbox is also available to toggle the visibility of the created traction vectors.

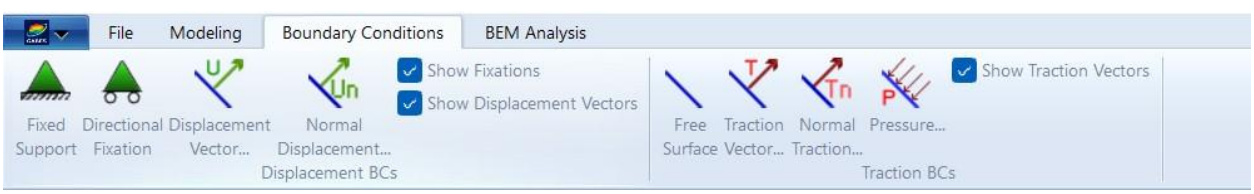
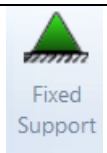
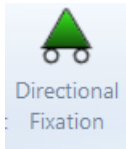
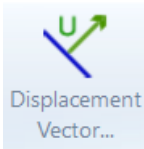
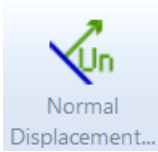
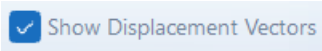
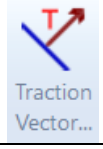
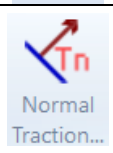
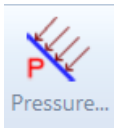
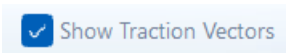


Figure 15: Boundary Conditions tab

These boundary condition assignment tools are only active when a **Surface Subset** is selected in the **Model Tree**. The commands on the **Boundary Conditions tab** are summarized in Table 5.

Table 5: Command buttons of the Boundary Conditions tab

Panel	Button	Action
Displacement BCs		Assigns a fully fixed boundary condition (zero displacement in all three directions) to the selected surface subset.
		Assigns zero displacement in the prescribed directions. The user is prompted with a dialog box to select the directions where zero displacement is applied.
		Assigns prescribed displacements in selected directions. The user is prompted with a dialog box to specify the directions and the corresponding displacement values.
		Assigns a displacement vector normal to the elements with a prescribed magnitude. The user is prompted with a dialog box to specify the displacement magnitude.
		Toggles the visibility of the created fixations on and off.
		Toggles the visibility of the created displacement vectors on and off.
Traction BCs		Removes the previously assigned boundary conditions from the selected surface, restoring it to a traction-free state.
		Assigns prescribed tractions in specified directions. A dialog box prompts the user to select the directions and enter the corresponding traction values.
		Assigns a traction vector normal to the elements with a specified magnitude. A dialog box prompts the user to enter the traction magnitude.
		This command is similar to the **Normal Traction** command but acts in the opposite normal direction. Positive pressure is applied inward.
		Toggles the visibility of the created traction vectors on and off.

3.3.4 BEM Analysis tab

The tab is divided into two panels:

- **Analysis Panel:** Contains commands for defining material properties, setting numerical integration parameters, managing memory, and launching simulations.
- **Post-Processing Panel:** Contains commands and tools for various post-processing tasks, including enabling or disabling the post-processing view mode; showing or hiding the deformed configuration of the body; setting the displacement magnification scale; selecting palette options; obtaining internal domain responses either at individual points or as tabulated reports for multiple points; and toggling the visibility of the maximum and minimum locations of the color-mapped field variables.

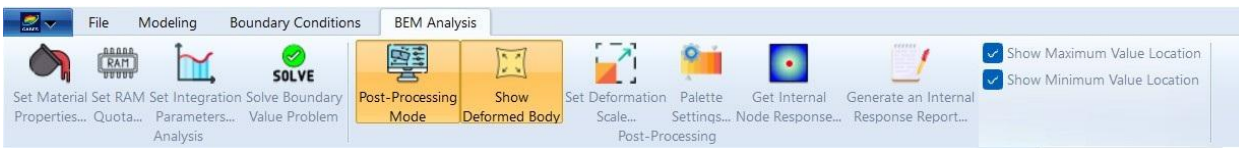
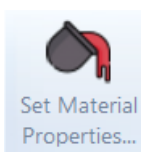
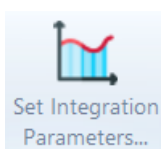
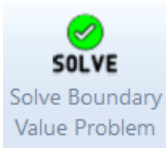
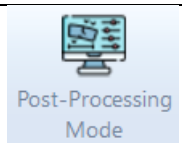
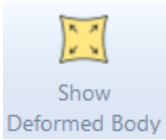
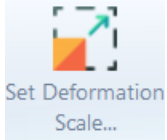
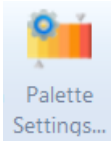
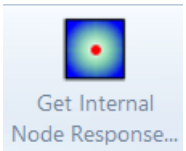
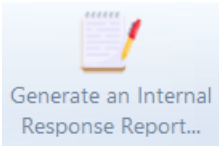


Figure 16: BEM Analysis tab

The commands on the **BEM Analysis tab** are summarized in Table 6

Table 6: Command buttons of the BEM Analysis tab

Panel	Button	Action
Analysis		Prompts the user with a dialog box to specify the material parameters. For linear elasticity, these include the Young's modulus E, the shear modulus μ, and the Poisson's ratio ν. By default, the parameters are set to the values for structural steel ($E = 200$ GPa, $\nu = 0.3$).
		Prompts the user with a dialog box to specify the reserved RAM memory quota for the BEM system matrix. If the matrix exceeds its allocated RAM quota, the remaining data is stored on the hard drive using the Out-Of-Core (OOC) storage method.
		This command opens a dialog box for configuring various integration parameters, including whether to use the Rigid Body method, selecting the cubature rule, or enabling an adaptive integration criterion.

		This command is the most important operation in the application. Once the model is fully prepared and all parameters are correctly defined, the user clicks this button to solve the problem. Before execution, a dialog box prompts the user to select various $\mathscr{H}$-matrix computation parameters and the solver to use (H-LU, BiCGStab, or GMRES).
Post-Processing		This button enables the Post-Processing View mode
		This button becomes active when the post-processing view mode is selected. After the problem is solved, this command allows the deformed body to be displayed by applying the computed displacement field to the mesh vertices using a uniform magnification scale.
		This button opens a dialog box for setting the deformation magnification scale. It becomes active only in post-processing view mode and after selecting the previous **Show Deformed Body** button
		Opens a dialog box for configuring the number formatting displayed in the Color Palette. Tick values can be shown in either floating-point or scientific (exponential) notation, with a user-defined number of decimal digits.
		Opens a dialog box for computing specified field variables at an internal domain point or on a boundary point. The user enters the spatial coordinates of the point and selects the variables to be evaluated (displacements, stresses, or strains). The results are displayed directly in the dialog box.
		This command serves the same purpose as the previous one. However, instead of specifying a single point, the user is prompted to select a point cloud (multiple points), which can be generated interactively or imported from an input coordinate file. After computation, the results are exported to a tabulated report file in CSV format, readable by spreadsheet programs such as Microsoft Excel.

	Show Maximum Value Location	Toggles the visibility of a small glyph, in the form of a yellow wireframe sphere, used to indicate the location of the maximum value on the 3D model.
	Show Minimum Value Location	Toggles the visibility of a small glyph, in the form of a magenta wireframe sphere, used to indicate the location of the minimum value on the 3D model

The View toolbar

The View toolbar is designed as a classical toolbar. Its purpose is to permit easy interaction, visualization, and manipulation of the 3D model. It offers the following functionalities:

- Changing the view angle (Face, Right, Top, Back, Left, Bottom, or Isometric).
- Changing the mesh display mode (Solid, Wireframe, or Edged Faces).
- Toggling the visibility of vertices and DOF nodes On and Off.
- Selecting the mouse action (View rotation, Element selection, Point selection, or None).
- Selecting the variables to display using a color-mapped distribution.

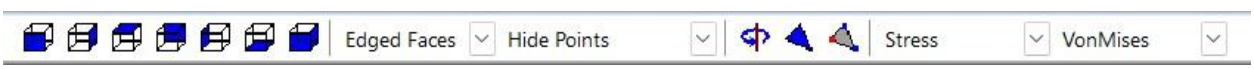
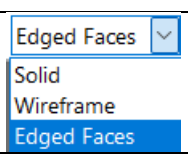
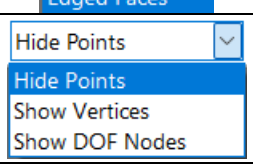
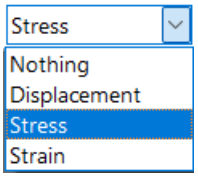
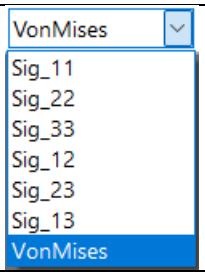


Figure 17: The View toolbar

The various tools and commands of the **View toolbar** are summarized in Table 7

Table 7: Functionalities of the View toolbar

Tool	Description
	These seven (07) buttons are used to change the view angle to specific well-known positions. The positions are, respectively: Face, Right, Top, Back, Left, Bottom, and Isometric
	This is the Output-Mode combo box, in which the user can modify the display mode of the mesh to one of the following: Solid, Wireframe, or Edged Faces.
	This combo box is used to show or hide the model vertices and DOF nodes. The user can choose one of the following: Hide Points, Show Vertices, or Show DOF Nodes.
	These three buttons define the current mouse interactive action. They serve, respectively: Rotate the model, Select elements, and Select points. At most one button can be selected at a time, or all of

	them can be deselected to prevent any interactive mouse action.
	This combo box is used in the post-processing phase to choose the field variable group, which can be: Displacement, Stress, Strain, or Nothing (if no variable color mapping is needed)
	This combo box lists the different variables available for color mapping distributions. Depending on the selected field variable group, the contents of this combo box change.

3.4 Dockable panels

Dockable panels are user interface components that can be attached to the edges of the main application window or displayed as independent floating windows, allowing users to customize the workspace layout according to their preferences and workflow. GABES has two distinct dockable panels, the Model-Tree panel and the Information panel. (See Figure 18).

3.4.1 The Model-Tree panel

The **Model Tree** is a floating and dockable panel that can be attached to any edge of the main frame window (see Figure 18). However, since the model hierarchy typically grows in the vertical direction, it is more practical to dock this panel on the **left or right side** of the main frame, where greater vertical space is available and navigation becomes more convenient.

The Model Tree panel displays the objects created during the modeling phase, including reference coordinate frames, the imported mesh, and the associated surface subsets. The panel is organized as a hierarchical tree structure composed of two (02) root nodes:

- **Reference Frames root**: contains all user-defined reference coordinate frames as well as the default global frame, which is permanently associated with the model's center of gravity and aligned with the center of the Client Area.
- **Mesh root**: becomes available once a mesh is loaded and adopts the name of the mesh file. All generated surface subsets are organized as child nodes of this root.

The Model Tree facilitates the modeling workflow by enabling easy modification of reference frames and straightforward assignment and tracking of boundary conditions

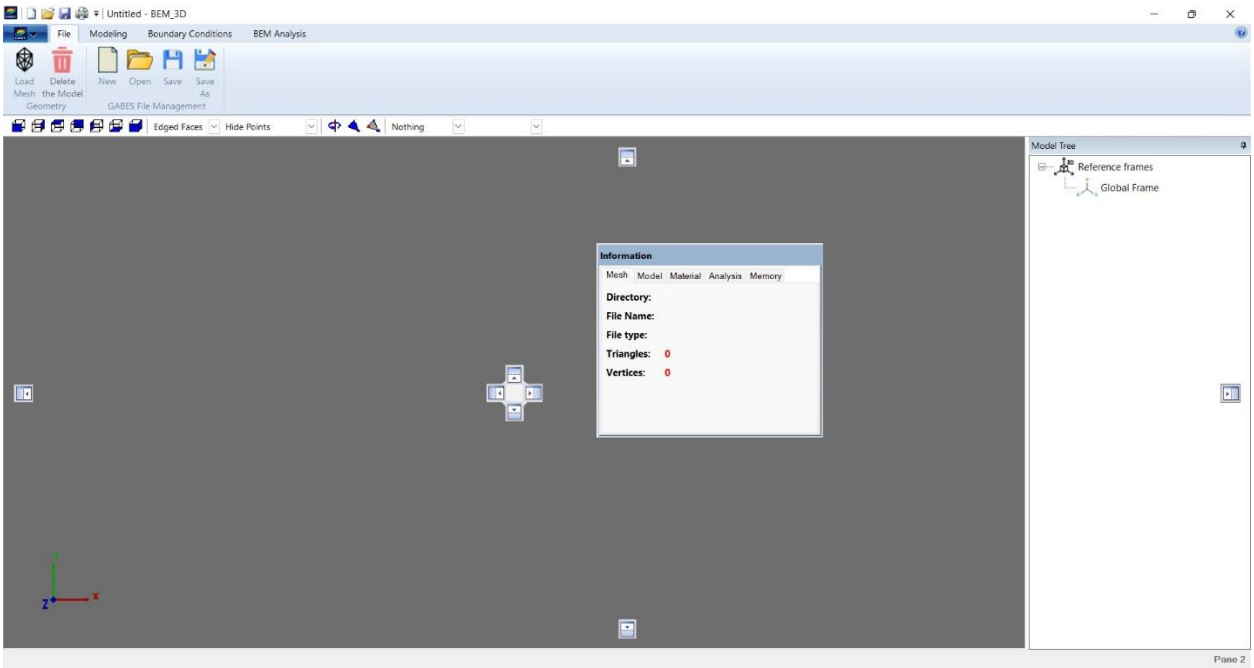


Figure 18: Dockable panels

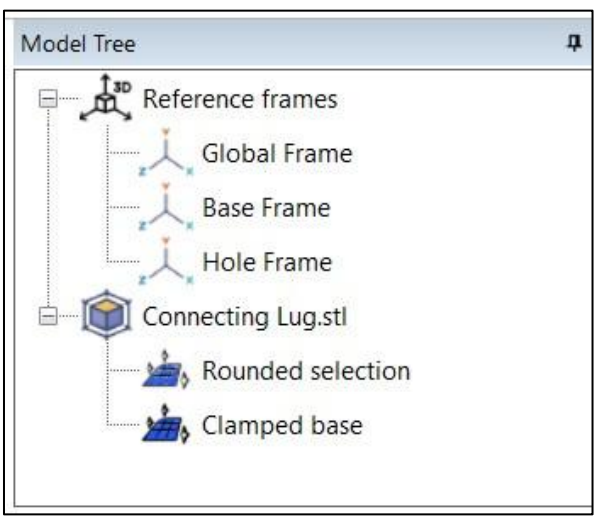


Figure 19: The Model-Tree panel

The objects displayed on the **Model-Tree panel** are summarized in Table 8

Table 8: The Model-Tree panel objects

Object	Description
Reference frames	Root node containing all reference coordinate frames defined in the model.
Global Frame	The global reference coordinate frame automatically created by the application and

	permanently associated with the model's center of gravity.
Hole Frame	A reference coordinate frame created by the user for custom positioning, orientation, or boundary-condition definition.
Connecting Lug.stl	Root node representing the imported mesh; its name is extracted from the mesh file title.
Clamped base	A surface subset created by the user from the loaded mesh, used for selection and boundary-condition assignment.

3.4.2 The Information panel

The **Information panel** is a floating and dockable panel that can be attached to any edge of the main frame window (see Figure 18). It is particularly convenient to dock it below the Model Tree, as this preserves more space for the main view area.

This panel shows essential geometrical and numerical information, along with memory storage requirements. This helps the user understand the size of the model and the system matrix, making it easier to adjust simulation settings efficiently.

As shown in Figure 20, the various information is displayed in five (05) separated tabs:

- Mesh tab
- Model tab
- Material tab
- Analysis tab
- Memory tab

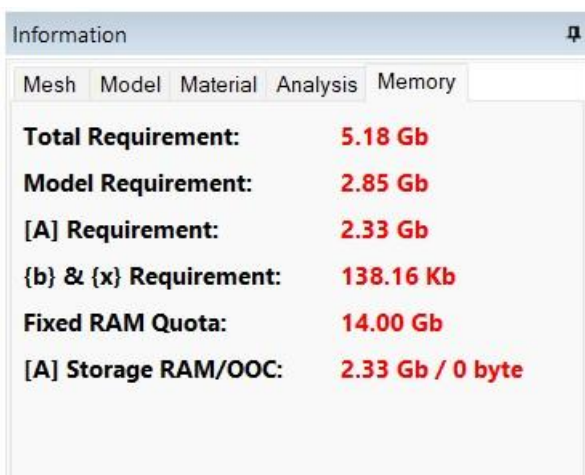


Figure 20: Infromation panel

3.4.2.1 Mesh tab

Displays mesh related information, like the location directory of the mesh file, the file title, the file type, the number of triangles and the number of vertices. (See Figure 21).

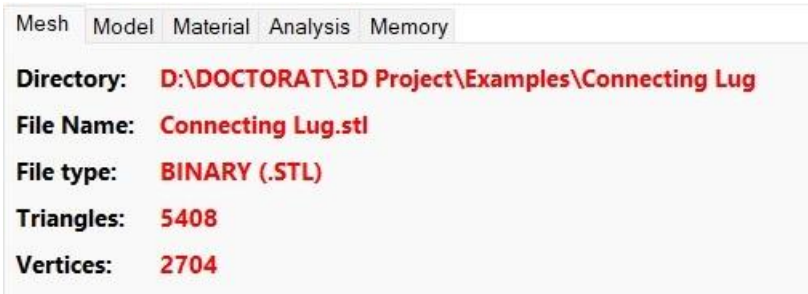


Figure 21: Information panel - Mesh tab

The information displayed under the **Mesh tab** is summarized in Table 9

Table 9: The information displayed under the Mesh tab

Information field	Description
Directory	The current hard disk directory where the mesh file is located.
File name	The name of the mesh file.
File type	The mesh file format, which can be either **STL** (binary or ASCII) or **OBJ** (ASCII).
Triangles	The total number of triangles in the mesh.
Vertices	The total number of vertices in the mesh.

3.4.2.2 Model tab

Displays model related information, like the discretization method, the number of DOF vertices which may differ from the number of the geometrical mesh vertices, the number of continuous and discontinuous elements and the resulting number of the algebraic DOFs which is three (03) times the number of DOF vertices due to the three (03) spatial directions. (See Figure 22)

Figure 22: Infromation panel - Model tab

The information displayed under the **Model tab** is summarized in Table 10

Table 10: The information displayed under the Model tab

Information field	Description
Disc. Type	The discretization method used, which can be Continuous, Discontinuous, or Disc. Sharp Edges. This corresponds to purely continuous elements, purely discontinuous elements, or automatically placed discontinuous elements at sharp edges and corners.
DOF Vertices	The number of degree-of-freedom (DOF) nodes, which may differ from the number of mesh vertices when discontinuous elements are present.
Cont. Elements	The total number of continuous elements in the model.
Disc. Elements	The total number of discontinuous elements in the model.
Algebraic DOFs	The resulting number of algebraic degrees of freedom, calculated as the number of DOF vertices multiplied by the three spatial dimensions. This value is important because it determines the overall size of the numerical system, including both matrix and vectors.

3.4.2.3 Material tab

The tab displays three material parameters: Young's modulus, shear modulus, and Poisson's ratio. In GABES, Young's and shear modulus are unitless, since the application operates in a dimensionless system. The actual units of traction and stress depend on the user's choice of length units when creating the mesh. For example:

- If geometry is interpreted in meters, traction is in Pascals (Pa).
- If geometry is in millimeters, traction is in Megapascals (MPa).

By default, the material values correspond to structural steel, and dimensions are assumed to be millimeters. If the user chooses a different unit system, the Young's modulus and shear modulus must be updated to remain consistent with the dimensionless system. This can be done via the **Set Material Properties…** button on the **BEM Analysis** tab of the ribbon.

N.B: Since Young's modulus and shear modulus are closely related, setting one is sufficient; the other will be computed automatically.

Figure 23: Information panel - Material tab

3.4.2.4 Analysis tab

This tab displays the matrix dimensions along with the current numerical integration parameters. It provides users with a clear overview of the system size and the specific integration settings applied during matrix construction, such as the method used to evaluate CPV integrals and whether an adaptive criterion scheme or a fixed cubature rule is employed.

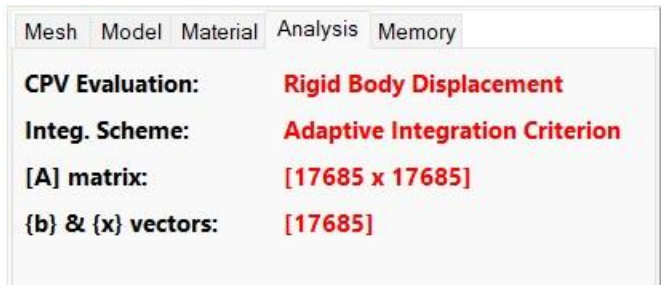


Figure 24: Information panel - Analysis tab

The information displayed under the **Analysis tab** is summarized in Table 11

Table 11: The information displayed under the Analysis tab

Information field	Description
CPV Evaluation	The method used to evaluate Cauchy Principal Value (CPV) integrals, which can be either **Rigid Body Displacement** or **Direct Evaluation**. This option can be modified using the **Set Integration Parameters…** button located under the **BEM Analysis** tab in the ribbon
Integ. Scheme	Specifies the integration scheme used to evaluate regular integrals, which can be either an **Adaptive Integration Criterion** or a **Fixed Cubature** with the corresponding number of integration points. This option can also be modified using the **Set Integration Parameters…** button.
[A] matrix	The dimension of the system matrix, which groups all influence coefficients associated either with the displacement U_{ij} kernel or the traction T_{ij} kernel. This matrix is always a square matrix.
{b} & {x} vectors	The dimension of the **{b}** and **{x}** vectors, which represent, respectively, the right-hand side vector and the unknown vector of the linear system $[A]\{x\} = \{b\}$.

3.4.2.5 Memory tab

GABES uses **$\mathcal{H}$-Matrix techniques** to accelerate the solution of the linear system but does not yet benefit from $\mathcal{H}$-Matrix memory compression. This approach simplifies the handling of boundary conditions by assembling a fully populated matrix. Once assembly is complete, the dense matrix is converted into an $\mathcal{H}$-Matrix format, enabling fast solvers while keeping the assembly algorithm simple.

Because a dense matrix is used during assembly, memory requirements can be significant. To manage this, GABES employs an **Out-of-Core** storage strategy, storing part of the data on disk to

avoid exceeding available RAM. As shown in Figure 25, the **Memory** tab displays various memory requirements, helping users determine how much RAM to allocate to the matrix. RAM allocation can be adjusted via the **Set RAM Quota…** button under the **BEM Analysis** tab of the ribbon. This approach allows simulation of relatively **large meshes** while keeping the matrix assembly algorithm simple.

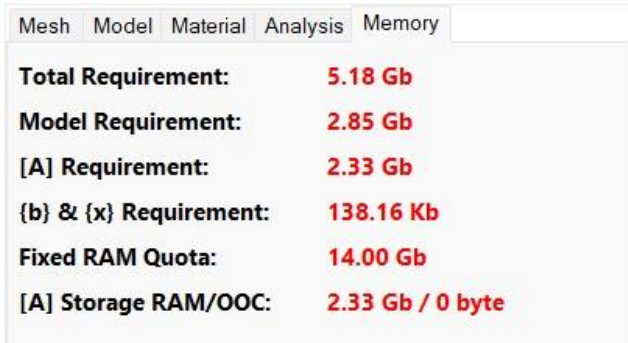


Figure 25: Information panel - Memory tab

The information displayed under the **Memory tab** is summarized in Table 12

Table 12: The information displayed under the Memory tab

Information field	**Description**
Total Requirement	The total amount of memory required for the simulation.
Model Requirement	The memory required for the internal representation of the mesh, including the BEM discretization data and the Direct3D structures used to render the model. **N.B.** The high memory requirement is due to the use of reusable integration data, which avoids recalculating quantities repeatedly during the numerical integration process. This method saves a considerable amount of time, but at the expense of relatively significant storage usage.
[A] Requirement	The total memory required to store the system matrix **A**. This typically represents the largest portion of memory usage, since the storage of a dense square matrix grows quadratically as $O(N^2)$ and can reach several gigabytes.
{b} & {x} Requirement	The total memory required to store each of the **b** and **x** vectors. This storage is generally negligible compared to the matrix and model memory requirements.
Fixed RAM Quota	The maximum amount of memory for the system matrix **A** allowed in Random Access Memory (RAM). This quota is defined by the user and cannot exceed the available system RAM
[A] Storage RAM/OOC	The actual storage distribution of the **A** matrix, showing the portion stored in RAM (left) and the portion stored on disk using Out-of-Core storage (right), separated by a slash (/).

3.5 The status bar

Status bars are user interface components supported by MFC and commonly included in application templates. They are widely used in desktop applications to display contextual command descriptions, application state information, and keyboard state indicators such as Caps Lock or Num Lock.

In GABES, the status bar is used exclusively to display the global coordinates of elements and vertices when they are hit-tested (i.e., when the mouse cursor passes over selectable geometry) (See Figure 26). This feature provides useful feedback during modeling operations, for example when creating coordinate frames and attaching them to well-defined locations.

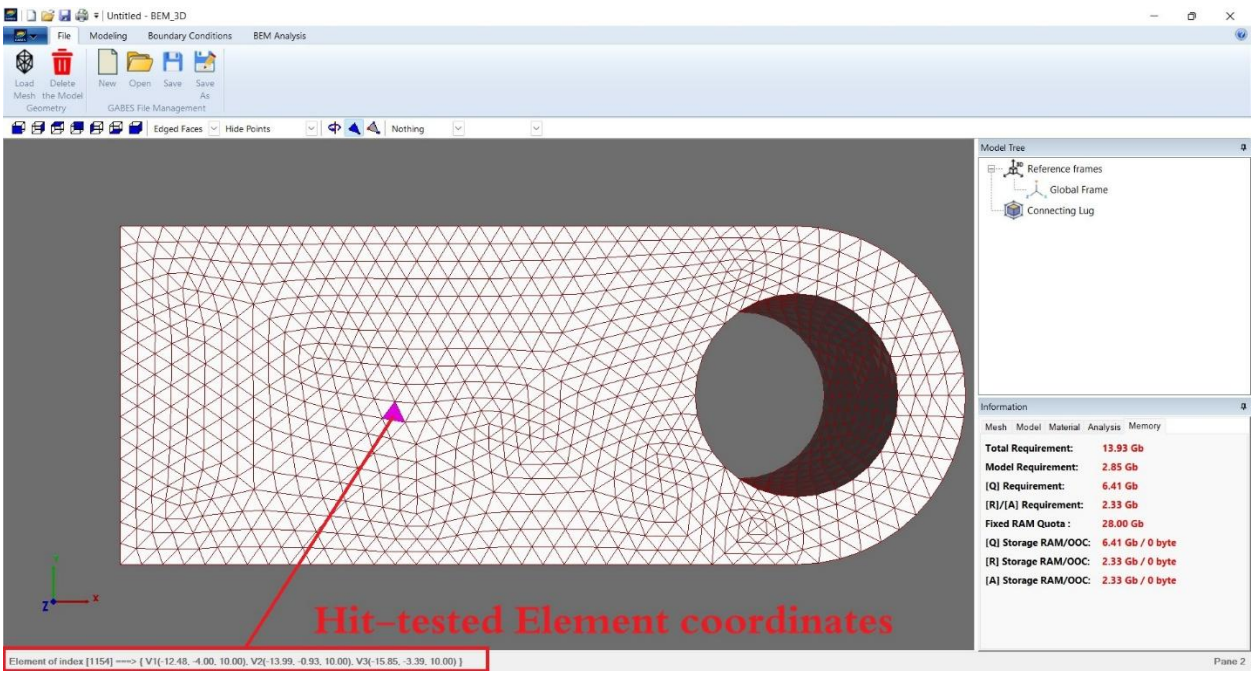


Figure 26: Status bar

4 USING GABES

4.1 Mesh creation

GABES does not implement internal geometry creation or mesh generation algorithms. Instead, it relies on established external software for these tasks. The application operates exclusively on triangular surface meshes, which are naturally suited to the Boundary Element Method (BEM) due to its boundary-only discretization approach.

GABES supports common mesh formats such as **Stereolithography (.stl)** and **Wavefront Object (.obj)**, both widely used in CAD, visualization, and multimedia applications.

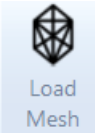
Polygonal modeling software, such as **Blender** and **Autodesk 3ds Max**, can directly produce compatible and well-structured meshes by exporting models in (.stl) or (.obj) format. In contrast, parametric CAD systems, such as **SolidWorks**, which are based on Boundary Representation (B-Rep) geometry, require conversion to triangular surface meshes before being used in GABES.

Although SolidWorks can export (.stl) files, this functionality is primarily intended for visualization or additive manufacturing and provides limited control over mesh quality and refinement. For better mesh control, SolidWorks models should either be exported as (.stl) files and then edited to form high-quality meshes in programs like **MeshLab** (open-source software), or exported to neutral exchange formats such as (.STEP) or (.IGES) and then meshed using **Gmsh**, an open-source mesh generator that allows mesh refinement and optimization. The Gmsh-generated meshes can then be exported in (.stl) or (.obj) formats compatible with GABES.

- **Gmsh** is available at: https://gmsh.info
- **MeshLab** is available at: https://www.meshlab.net

4.2 Mesh loading

Once a good-quality mesh is created and exported in **(.stl)** or **(.obj)** formats, it is recommended to store it in a dedicated folder. When the mesh is loaded into GABES, its directory automatically becomes the application's working directory, and all GABES generated files are saved there by default. Using separate folders for each model helps keep project files organized and easy to manage.



A mesh is loaded using the **Load Mesh** button situated in the **File** tab of the ribbon. When this button is clicked, an **Open File** dialog box will appear, allowing the user to browse and select the desired mesh file. Before the mesh is fully loaded, the **Set Mesh Type** dialog box appears, prompting the user to choose the discretization method to be applied to the mesh. In this context, discretization refers to the boundary element type assigned to each triangular mesh element.

GABES uses two types of linear elements: continuous and discontinuous elements. Both element types have three (03) degrees of freedom (DOF) nodes, differing only in their location, as illustrated in Figure 27.

Continuous element nodes coincide with the three geometric vertices of each triangular element, resulting in node sharing between neighboring triangles. In contrast, discontinuous element nodes are located inside the triangle, creating independent degrees of freedom (DOFs) for each element.

Continuous elements, due to node sharing, significantly reduce the overall number of degrees of freedom. However, they tend to yield poor results at sharp edges and corners because of the abrupt variation of the outward normal vector between adjacent elements. On the other hand, discontinuous elements provide better accuracy near edges and corners, and the assembly of the numerical linear system is simpler because no DOF reduction is required due to the absence of

node sharing. Their main drawback is the generation of larger linear systems with a higher number of degrees of freedom, which can negatively affect computational performance.

GABES addresses this compromise by automatically assigning continuous elements to smooth boundaries while enforcing discontinuous elements at sharp edges and corners where the angle between adjacent element normals exceeds 45°. This automatic placement of discontinuous elements in sharp regions is the default behavior in GABES. However, this behavior can be overridden by selecting a uniform element type through the **Set Mesh Type** dialog box, which appears before loading a mesh and offers three options: **Purely Continuous**, **Purely Discontinuous**, or **Discontinuous at Sharp Edges**. As shown in Figure 28.

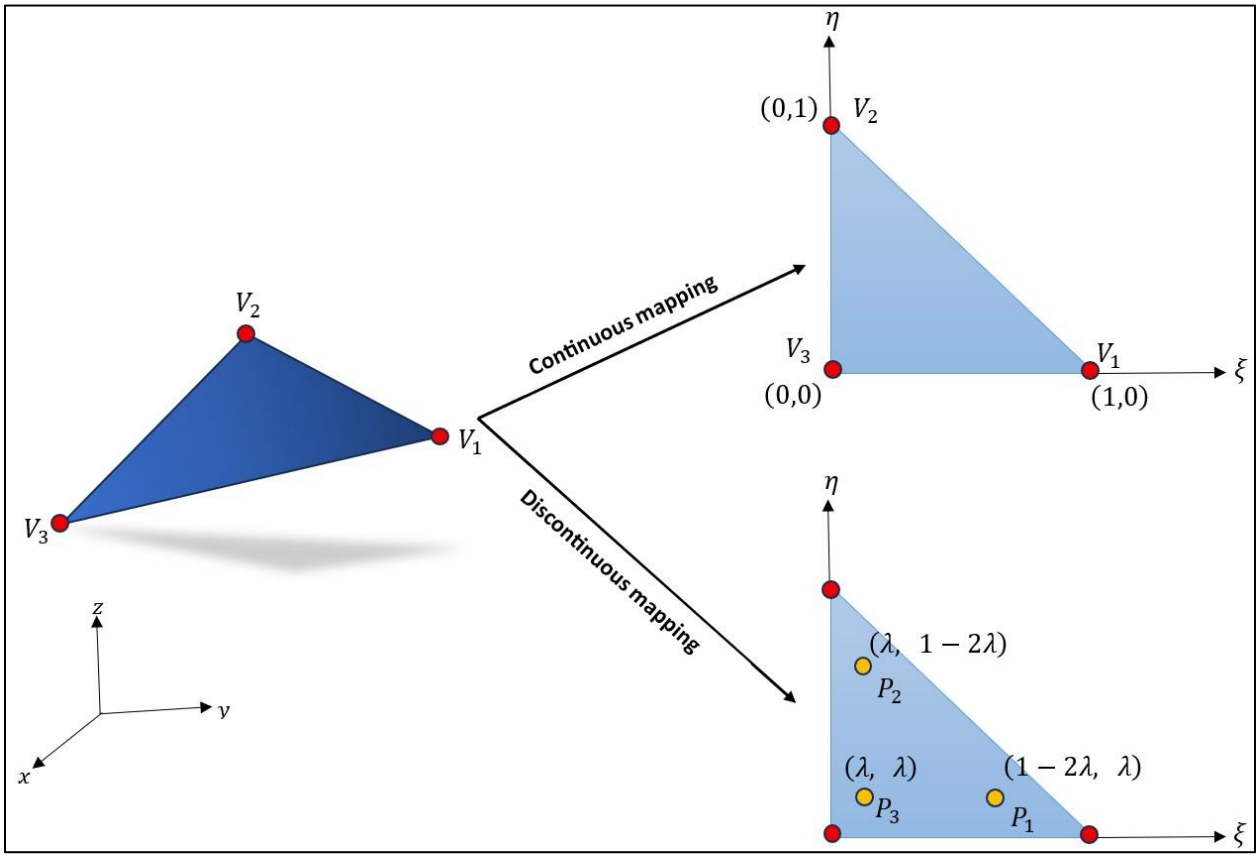


Figure 27: Boundary Element Types

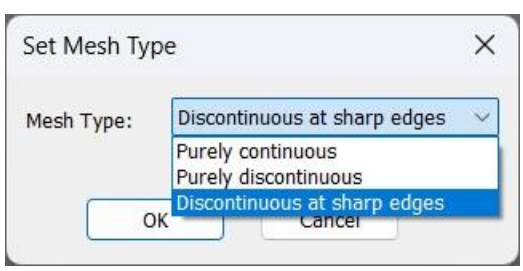


Figure 28: Set Mesh Type dialog box

4.3 Modeling operations

Modeling operations indicate all the actions performed by the user to prepare the model for simulation. In the context of GABES, modeling operations generally consist of two main tasks:

- Reference frames definition
- Surface subsets creation

4.3.1 Reference frames definition

Reference frames are orthogonal, right-handed coordinate systems that include the default global frame as well as any custom frames created by the user. Their primary function is to organize and interconvert the coordinates of three-dimensional objects, such as vertices, nodes, and elements. Reference frames are especially useful for facilitating the interactive creation of surface subsets and for providing intuitive coordinate mapping during post-processing. By allowing the user to define points relative to well-defined geometrical locations, reference frames simplify the interpretation and input of coordinates.

This functionality is particularly important because each loaded model is automatically centered on the screen, with all its coordinates mapped to the global frame based on the model's center of gravity. As a result, the raw global coordinates may not be intuitive for the user. Custom reference frames enable more meaningful and user-friendly coordinate systems, making modeling, selection, and post-processing operations easier and more precise.

4.3.1.1 Create a new Reference Frame

To add a new reference frame, navigate to the **Modeling** tab in the ribbon and select the **Add Reference Frame…** button. The **Create/Modify Reference Frame** dialog box shown in Figure 29 will appear. This is a **modeless dialog**, which means that the main application window remains active, allowing the user to interact with the model, such as selecting vertices and elements graphically, while defining the new reference frame.

4.3.1.2 Modify an existing Reference Frame

The **Create/Modify Reference Frame** dialog box can also be used to modify an existing coordinate frame, except for the default global frame, which is fixed and tied to the model's center of gravity. A user-defined frame can be modified by simply double-clicking it in the **Model Tree** panel. This action opens the **Create/Modify Reference Frame** dialog box, allowing the user to change the frame's location or orientation.

4.3.1.3 The use of the Create/Modify Reference Frame dialog

This dialog box, shown in Figure 29, allows the creation or modification of right-handed orthogonal coordinate systems using a simplified approach. Normally, defining a 3D coordinate system requires six quantities: the coordinates of the origin point and three Euler angles. However, Euler angles can be difficult for users to visualize. To simplify the process, GABES does not require Euler angles. Instead, the user only needs to define a single reference plane (**OXY, OYZ,**

or **OZX**) by specifying two orthogonal unit vectors with a zero-dot product. The third unit vector is automatically computed using the vector cross product, ensuring a valid orthogonal right-handed coordinate system.

The procedure for defining a reference frame is as follows:

1. Define the origin point.
2. Select a reference plane using the **Plane** dropdown combo box.
3. Assign the unit vectors for the axes of the chosen plane.

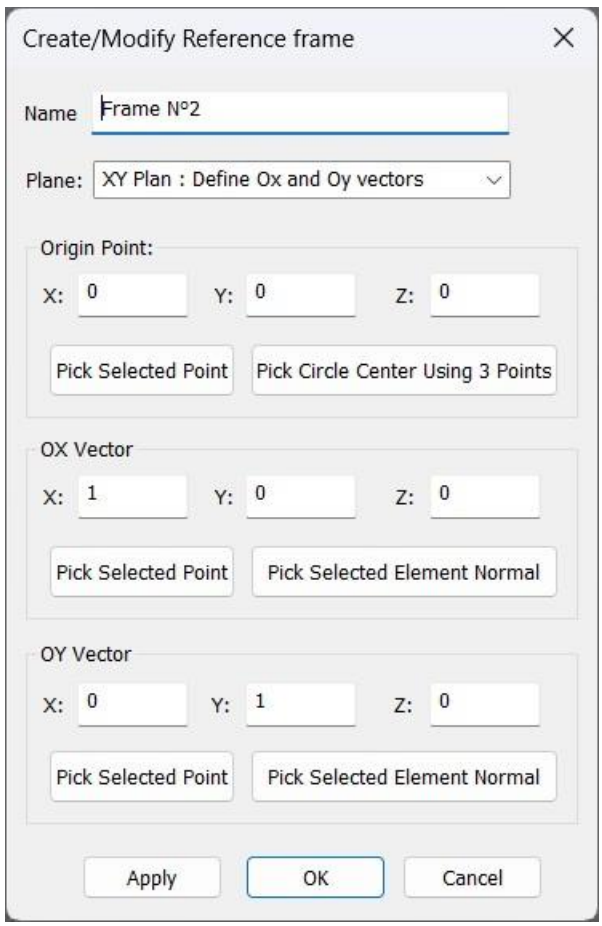


Figure 29: Create/Modify Reference Frame

The origin point can be defined either manually or graphically:

a. **Manual Input:** Enter the desired coordinates directly, expressed in the global coordinate system.

b. **Graphical Methods:**

- **Pick Selected Point:** Use this method to place the frame on a known vertex, such as a corner or edge tip. It requires a prior and valid graphical selection of a single point; clicking the **Pick Selected Point** button places the origin of the frame at that selected point.
- **Pick Circle Center Using 3 Points:** Use this method to place the frame at the center of a circular edge. This is useful, for example, for defining a cylindrical coordinate mapping with the Z-axis aligned along the cylinder axis. It requires a valid graphical selection of three distinct points on the circular edge; clicking the **Pick Circle Center Using 3 Points** button places the frame origin at the circle center.

The unit vectors can also be defined either manually or graphically:

a. **Manual Input:**
Enter the three (03) vector components directly.

b. **Graphical Methods:**

- **Pick Selected Point:** Use this method to orient the vector from the origin toward a graphically selected point, such as a corner or an edge tip. It requires a prior valid graphical selection of a single point. Clicking this button creates a unit vector oriented from the origin toward the selected point.
- **Pick Element Normal:** Use this method to orient the vector along the outward normal direction of a selected element. It requires a prior valid selection of a single element. Clicking this button creates a unit vector perpendicular to the selected element and oriented outward. This method is useful for defining a vector normal to a known surface by simply selecting one of its elements.

It is worth noting that all created reference frames are displayed using a small gizmo, allowing users to identify their exact location without obscuring or interfering with the 3D model. The gizmo axes are temporarily scaled when a frame is selected, during frame creation or modification, as well as during surface subset creation, to improve visibility and facilitate interactive operations.

Reference frame gizmos are not displayed in the post-processing view mode, except when defining track nodes for internal response calculations or generating tabulated results. In these cases, the selected frames gizmos are shown scaled to enable an intuitive and interactive track node creation process.

Figure 30 below illustrates the appearance of reference frame gizmos for both a selected and unselected frame.

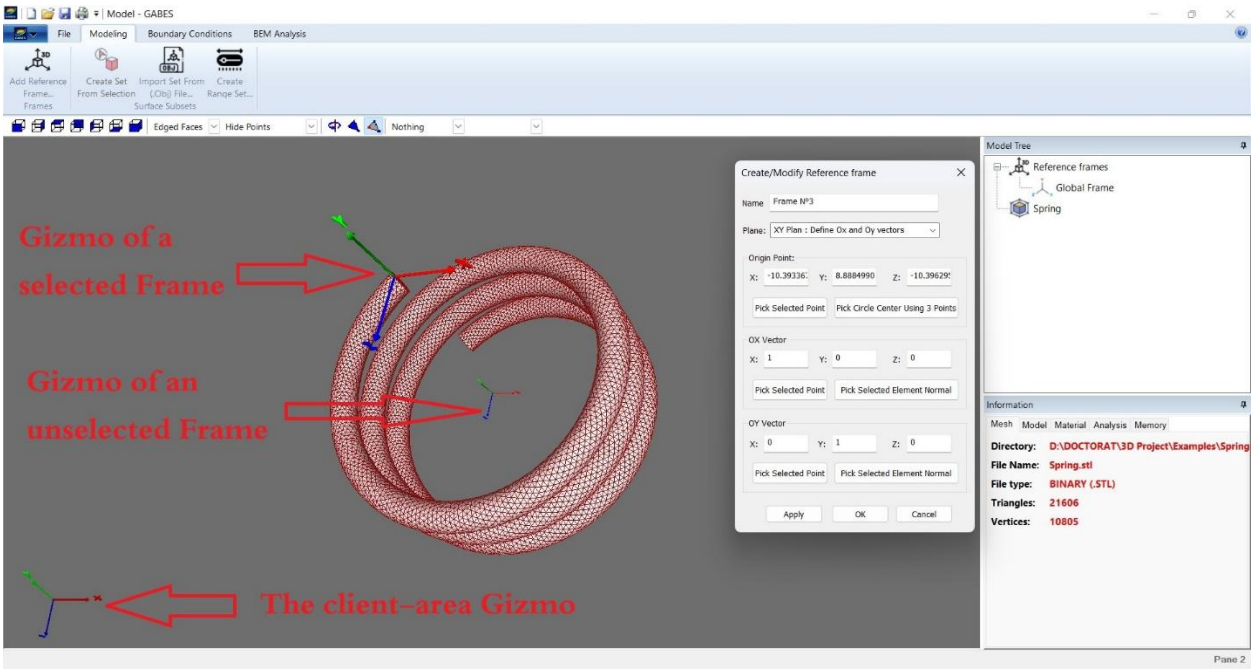


Figure 30: Reference Frame gizmos for selected and unselected frames

4.3.2 Surface subsets creation

As their name indicates, **surface subsets** are user-defined portions of the boundary and are mandatory for boundary condition assignment. They provide a flexible and efficient method for defining boundary conditions. All boundary condition tools located under the **Boundary Conditions** tab of the ribbon are enabled only after a valid surface subset is selected in the **Model Tree** panel.

GABES provides three (03) methods for creating surface subsets:

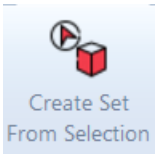
- Graphical Selection
- Import from an (.obj) file
- Interactive Range Selection

All these three methods command buttons are accessible under the **Modeling tab** of the ribbon

4.3.2.1 Graphical Selection

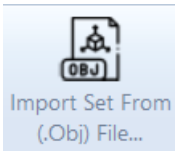
The user can select multiple elements after enabling the **Element Selection** button on the **View** toolbar. To select multiple elements, the **CTRL** key must be held while selecting.

When a valid element selection exists, the **Create Set From Selection** button becomes enabled. Clicking this button automatically converts the current selection into a surface subset with an automatically generated name, which appears in the **Model Tree** under the mesh root. The newly created subset can be renamed directly within the **Model Tree** panel.



4.3.2.2 Import from an (.obj) file

This method can be used with meshes created in **Autodesk 3ds Max**, which provides the ability to create sub-surfaces of the original model using its **Detach** command. These detached surfaces can be exported individually as **Wavefront (.obj)** files. GABES can then convert these detached surfaces into boundary subsets using the **Import Set From (.obj) File** command.



Clicking this button opens a dialog box that allows the user to browse and select the desired file. Once the subset is created, GABES automatically assigns it a name based on the filename of the imported surface.

When using this method, it is preferable to store the detached surface **(.obj)** file in the same directory as the mesh file, which serves as the default working directory for the model.

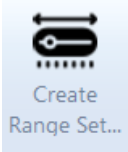
This method is commonly used in the **BESLE** BEM open-source software available at: https://github.com/GalvisA1087/BESLE_v2.0

4.3.2.3 Interactive range selection

This method is the most practical and versatile among the three. It relies extensively on reference frames, allowing elements to be selected using coordinate ranges defined in cartesian, cylindrical, or spherical coordinate systems. By placing user-defined reference frames at key geometrical locations on the 3D model, the user can easily determine the coordinate ranges required for selection.

For example, to select a surface with known dimensions, the user can place a reference frame at one corner of the surface and orient its axes so that the selection ranges can be conveniently defined using the X, Y, and Z coordinates. Another example involves cylindrical surfaces, where the user can place a reference frame at the center of the circular base and orient the Z-axis along the cylinder height. This configuration allows cylindrical regions to be easily selected by specifying ranges for the radius, angle, and height. A similar approach can also be applied to spherical geometries.

In addition to manual input, range selection can be performed interactively using slider track bars that define the minimum and maximum range values. The selection is updated graphically as the sliders are adjusted. When entering range values manually, the selection can be refreshed by clicking the **Apply** button.



Range-based subsets are created using the **Create Range Subset…** button, which opens the **Create Range Subset** modeless dialog box, as shown in Figure 31. The ranges are defined by selecting a coordinate frame from the **Coordinate Frame** drop-down list, which highlights and scales the frame gizmo to improve visibility in the 3D scene. The coordinate system can then be selected (Cartesian, Cylindrical, or Spherical), which updates the minimum and maximum values for each dimension. The ranges can be adjusted interactively using the slider bars or by manually entering values and updating the selection using

the **Apply** button. The dialog also allows the user to specify the subset name. Clicking **OK** converts the created selection into a surface subset which will appear in the **Model-Tree** panel.

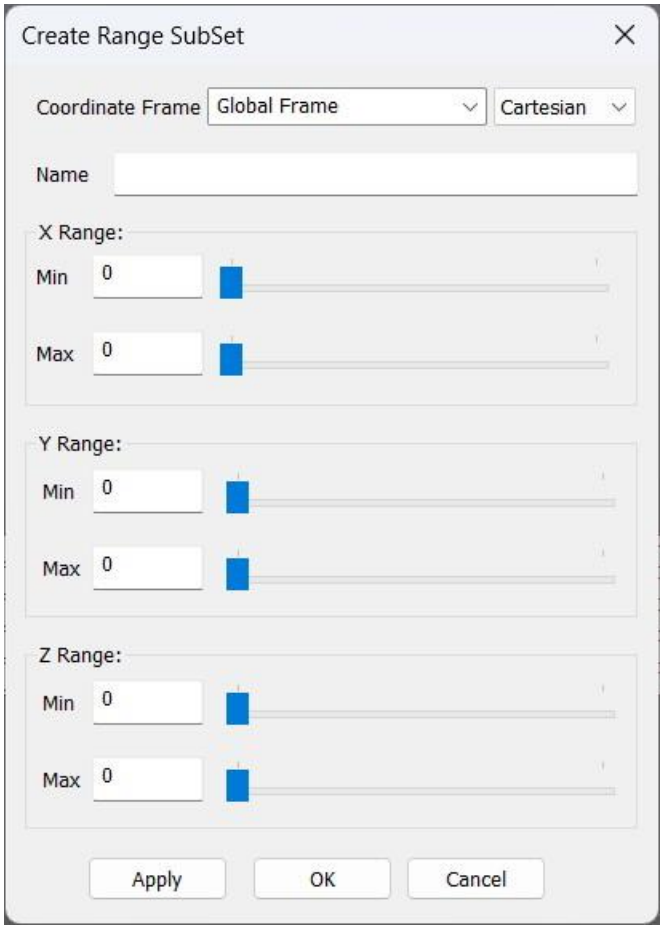


Figure 31: Create Range Subset dialog

Note:
The coordinate range extents are updated automatically behind the scenes after any modification to the mesh or after adding or modifying a coordinate frame. Each frame keeps track of the model's coordinate extents in its own local system. As a result, the ranges of the slider bars will exactly cover the full model range. The only exception is the angular ranges, which cover the complete angular extent regardless of the model's actual angular range.

4.4 Boundary conditions assignment

As stated in the above sections, the presence of the surface subsets is mandatory for boundary condition assignment. Once, the needed subsets are defined the application of boundary condition to them can be performed by using two (02) distinct approaches:

- The first approach is by using the **Subset Boundary Conditions** dialog box

- The second approach is by using the various boundary condition assignment tools available under the **Boundary Conditions** tab of the ribbon

4.4.1 The use of the Subset Boundary Conditions dialog box

This dialog box is invoked by double-clicking the corresponding subset icon in the **Model Tree** panel. As shown in Figure 32, it allows the user to rename the selected surface subset, verify and keep track of its current boundary condition values, and define its overall boundary state by specifying either displacement (Dirichlet) conditions or traction (Neumann) conditions. These conditions may be mixed across the three directions; for example, the user may prescribe traction in one direction and displacement in another.

In addition, the dialog box allows the definition of normal displacement or normal traction vectors. When a normal vector condition check box is selected, all other controls are disabled except for the corresponding normal magnitude field.

For vector components, when a fixed displacement component check box is selected, the associated displacement edit box is enabled and the corresponding traction edit box is disabled. When the check box is unchecked, the opposite occurs: the traction edit box becomes enabled and the displacement edit box is disabled.

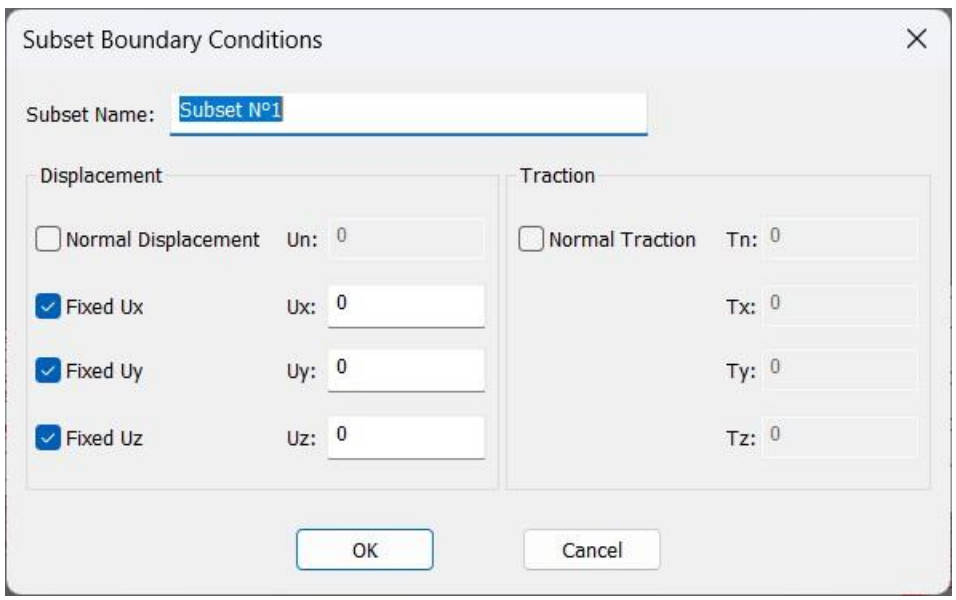
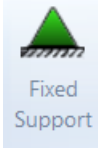


Figure 32: The Subset Boundary Conditions dialog box

4.4.2 The use of the Boundary Conditions tools of the ribbon

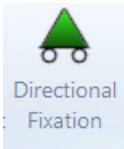
After the selection of a subset from the **Model-Tree**, the various boundary condition assignment buttons located under the **Boundary Conditions** tab become activated.

4.4.2.1 Fixed Support



This condition is applied using the **Fixed Support** button, which imposes zero displacement in all three spatial directions $U_1 = U_2 = U_3 = 0$.

4.4.2.2 Directional Fixation



This condition is applied using the **Directional Fixation** button, which imposes zero displacement along the selected directions. Clicking this button opens the dialog box shown in Figure 33, where the user can select the directions to be restrained.

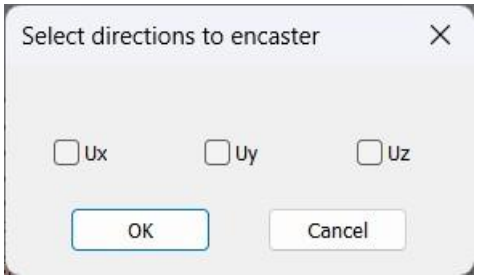
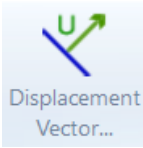


Figure 33: Select directions to encaster

4.4.2.3 Displacement Vector



This tool allows the user to apply specific displacements in chosen directions. Unlike the **Directional Fixation** tool, which restricts movement, this tool imposes a non-zero displacement. To use it, click the **Displacement Vector…** button to open the dialog box shown in Figure 34. Then, check the desired displacement component to enable its edit box and enter the required value.

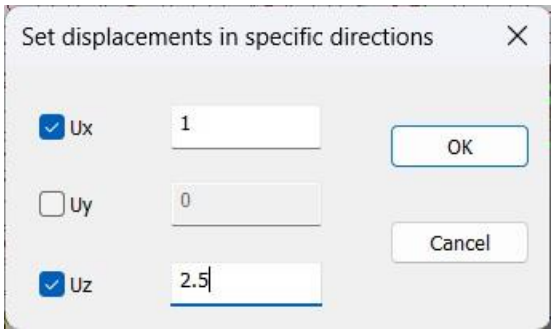


Figure 34: Set Displacements in specific directions

4.4.2.4 Normal Displacement

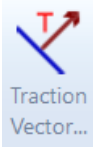
This tool allows the application of a non-zero displacement along the normal direction of the elements. It is assigned using the **Normal Displacement…** button, which opens the dialog box shown in Figure 35, where the displacement magnitude can be specified. A positive value applies the displacement in the outward normal direction, while a negative value applies it in the inward normal direction.

Figure 35: Set the normal displacement magnitude

4.4.2.5 Free Surface

This command removes all applied boundary conditions and restores the default free-surface condition, characterized by unknown displacement and zero traction. It is assigned by using the **Free Surface** button.

4.4.2.6 Traction Vector



This tool is similar to the Displacement Vector tool; however, it applies traction components in the selected directions. To use it, click the **Traction Vector…** button to open the dialog box shown in Figure 36. Then, select the desired traction component to enable its edit box and enter the required value.

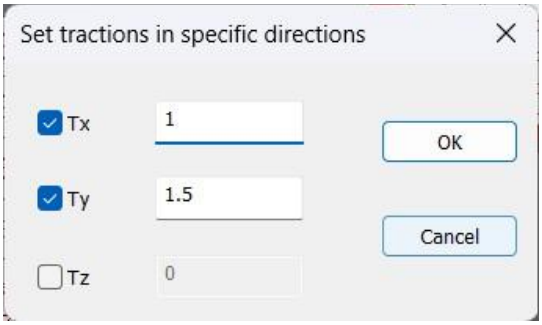
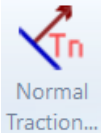


Figure 36: Set tractions in specific directions

4.4.2.7 Normal Traction



This tool applies traction along the element normal direction. It is accessed through the **Normal Traction…** button, which opens the dialog box shown in Figure 37, allowing the user to specify the traction magnitude. Positive values apply traction in the outward normal direction, whereas negative values apply it in the inward normal direction.

Figure 37: Set Normal traction

4.4.2.8 Pressure

This tool is equivalent to the **Normal Traction** tool but uses the opposite sign convention. In this case, a positive pressure value represents a load acting toward the surface, while a negative value represents a load acting away from the surface.

4.5 Analysis phase

Once the mesh is loaded, and the boundary conditions assigned, the boundary value problem is fully defined and the model become ready for simulation. However, some parameters need to be verified and set by the user before solving the boundary value problem.

4.5.1 Setting options before solving

4.5.1.1 The material properties

In linear elasticity, a material is characterized by Young's modulus E and Poisson's ratio ν. In some cases, the shear modulus μ is used instead of Young's modulus. These elastic material properties can be defined using the **Set Material Properties…** button, which opens the corresponding dialog box shown in Figure 38.

By default, the material is configured as structural steel with a Young's modulus of 200,000 MPa, assuming that dimensions are expressed in millimeters. The user may modify these properties to match the specific material under study.

The user must pay careful attention to the length unit used for the mesh vertices. GABES operates internally in a dimensionless unit system; therefore, if lengths are expressed in millimeters, tractions are in megapascals, whereas if lengths are expressed in meters, tractions are in pascals. Consequently, the Young's modulus or shear modulus must be defined consistently with the chosen unit system.

Since Young's modulus E and shear modulus μ are related by

$$\mu = \frac{E}{2(1+\nu)}$$

only one of them needs to be specified; the other is updated automatically

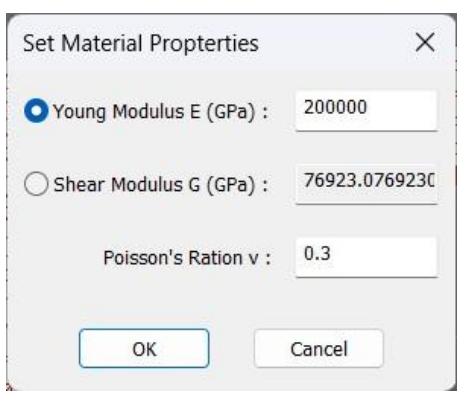
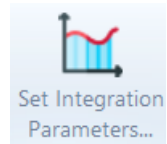


Figure 38: Set Material Properties

4.5.1.2 The numerical integration parameters



Some numerical integration parameters can be defined by the user. These include the method used to evaluate Cauchy Principal Value (CPV) integrals, which can be computed either directly or using the Rigid Body Displacement approach. The user must also specify the integration scheme for evaluating regular integrals, either through an adaptive criterion or a fixed cubature order. These options can be configured using the **Set Integration Parameters…** button, which opens the dialog box shown in Figure 39.

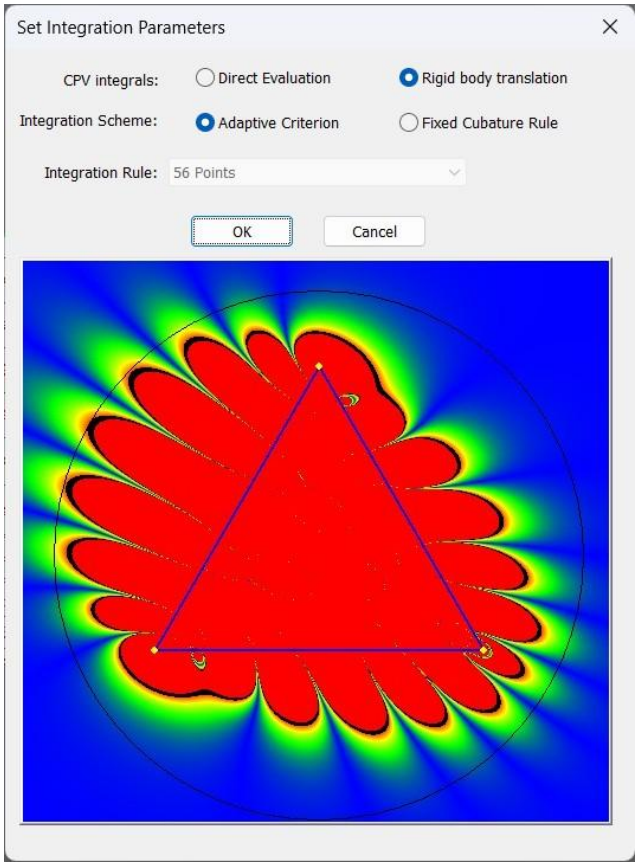


Figure 39: Set Integration Parameters

Note:
In the present version of GABES, the direct CPV evaluation is not yet fully and correctly implemented. The obtained results are generally of poor quality, particularly for discontinuous elements, as the method currently lacks sufficient numerical accuracy and stability. This limitation may affect the correctness of the results.

Therefore, it is strongly recommended to use the Rigid Body Displacement method instead.

4.5.1.3 The RAM Quota Allocation

Although the $\mathscr{H}$-Matrix framework enables matrix compression and can reduce storage requirements to nearly linear complexity, typically $O(N\log N)$, instead of the quadratic growth

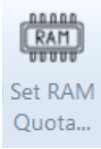
$O(N^2)$associated with dense matrices, GABES currently exploits the $\mathscr{H}$-Matrix technique only to reduce the solution complexity (factorization and solve stages).

This design choice is primarily related to implementation constraints rather than a theoretical limitation of the $\mathscr{H}$-Matrix framework. In principle, $\mathscr{H}$-Matrix can be constructed without explicitly assembling the full dense matrix. However, doing so requires specialized assembly algorithms capable of directly generating low-rank block representations from kernel evaluations. Such approaches are significantly more complex to implement than classical dense assembly strategies.

In the current architecture of GABES, matrix assembly relies on column-wise construction and column swapping strategies, which are naturally suited to a fully explicit dense matrix representation. Furthermore, the recommended Rigid Body Displacement (or Rigid Body Translation) method requires explicit access to off-diagonal block entries during assembly. In contrast, within an $\mathscr{H}$-Matrix representation, admissible off-diagonal blocks are stored in low-rank compressed form, meaning their full explicit entries are not directly available without reconstruction. This creates a practical incompatibility between the current assembly strategy and a purely direct $\mathscr{H}$-Matrix construction approach.

Consequently, the intermediate step of assembling the dense matrix may result in very high memory requirements for large-scale problems.

To mitigate this issue, GABES employs an Out-of-Core (OOC) storage strategy. For large systems, only a portion of the matrix is stored in RAM, while the remaining portion is temporarily stored on disk. The maximum amount of RAM that the matrix is allowed to occupy is defined in GABES by the **RAM Quota** parameter.



The RAM Quota can be configured by the user through the **Set RAM Quota** dialog shown in Figure 40, which is accessed by clicking the ***Set RAM Quota…*** button.

This dialog displays the total available physical memory of the host system and allows the user to define the RAM Quota using a slider bar. The slider range spans the available physical memory, and its ticks are expressed in gigabyte units.

The dialog also provides an automatic configuration option. In this mode, the application determines the RAM Quota by taking the total available physical memory, subtracting the memory required by the model and the system vectors, and assigning half of the remaining space to the matrix storage.

If the user chooses to set the RAM Quota manually, the memory requirements of both the model and the vectors must be considered. These values are available in the **Memory** tab of the Information panel. The user should also account for memory consumed by other processes running concurrently on the host system.

Finally, it is important to remember that disk-based Out-of-Core (OOC) storage is a high-latency memory medium and is significantly slower than system RAM. Therefore, selecting the RAM Quota involves a trade-off between available system resources and overall program performance.

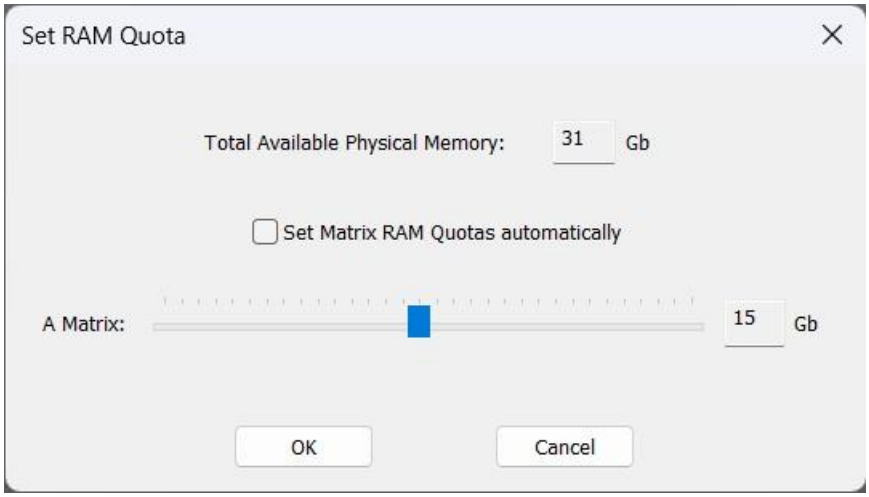
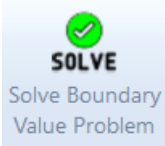


Figure 40: Set RAM Quota dialog box

4.5.2 Solving the linear system



Once the boundary value problem is fully defined and all associated numerical and physical parameters are correctly set, the solution process can be initiated.

The process begins by clicking the **Solve Boundary Value Problem** button. This opens the **Solver Parameters** dialog box (Figure 41), where the user can specify several essential $\mathscr{H}$-Matrix numerical parameters, summarized in Table 13.

After clicking **OK** in the Solver Parameters dialog, the calculation starts and the **Wait Lengthy Job** dialog box (Figure 42) appears. This dialog shows the overall progress of the operation, continuously reports the current background task, and displays the elapsed time.

The computation time can range from a few seconds to several minutes, depending on the model size, the allocated RAM Quota, and the selected solver parameters described in Table 13.

Table 13: Solver parameters and their descriptions

Parameter	Description
Solver	This parameter can be defined using the corresponding combo box to choose the solver method used for resolution, this can be either (H-LU, GMRes or BiCGStab). • H-LU: is a direct method which is the $\mathscr{H}$-Matrix version of the LU factorization • GMRes and BiCGStab: are two Krylov-subspace (iterative) solvers

Error threshold	This parameter defines the relative error, which serves as the threshold during the construction of the $\mathcal{H}$-Matrix approximation. The same threshold is also used when a Krylov-subspace (iterative) solver, such as GMRes or BiCGStab, is selected, controlling the convergence criterion of the iterative method.
Number of steps	This parameter specifies the maximum number of iterations and is only active when a Krylov-subspace solver (GMRes or BiCGStab) is selected. It is automatically disabled when the H-LU direct solver is chosen.
Minimum Leaf Size	This parameter defines the minimum size of the smallest $\mathcal{H}$-Matrix block (leaf size). Using a smaller minimum leaf size generally increases the compression of the matrix, but it also increases computation time. For typical 3D BEM elasticity problems, a minimum leaf size of 64 provides a good balance between compression and performance.
Maximum Rank	The $\mathcal{H}$-Matrix method uses low-rank representations to approximate far-field blocks. The maximum rank parameter specifies the highest allowed rank during the $\mathcal{H}$-Matrix approximation when using the Adaptive Cross Approximation (ACA) algorithm.

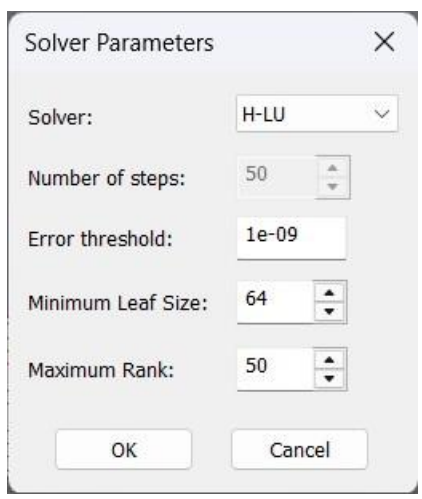


Figure 41: Solver parameters

Figure 42: The Wait Lengthy Job dialog box

4.6 The post-processing phase

When the boundary value problem solution completes successfully, the **OK** button in the **Wait Lengthy Job** dialog box becomes enabled. Clicking this button automatically activates the **post-processing view mode**, which by default displays the deformed configuration of the model along with the surface color-mapped Von-Mises stress.

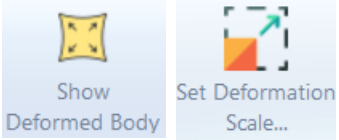
During the post-processing phase, the user can analyze the results in several ways, including:

- Viewing color-mapped distributions of various field variables.
- Calculating the response at any point on the boundary or within the domain.
- Generating detailed tabulated reports in **.csv** format, compatible with spreadsheet software such as Microsoft Excel.

4.6.1 The deformed configuration of the body

When the problem is solved, the displacement vectors of all nodes become known. Applying these displacements to the mesh vertices produces the deformed configuration of the body. In most cases, particularly within the elastic range, the magnitude of these displacements is very small. Therefore, as is common in engineering analysis applications, the deformations are magnified using a uniform scaling factor applied to all displacement vectors to make them visually perceptible.

In GABES, the deformed configuration of the model is displayed using the **Show Deformed Body** button. The deformation scale can be adjusted via the **Set Deformation Scale…** button, which opens the **Set Deformation Scale** dialog box, shown in Figure 43.



By default, the deformation scale is automatically computed so that the maximum deformation magnitude corresponds to ten (10) units in the 3D scene. The user can either enter a uniform scale manually or select the default auto-computation method by checking the **Auto Compute** checkbox.

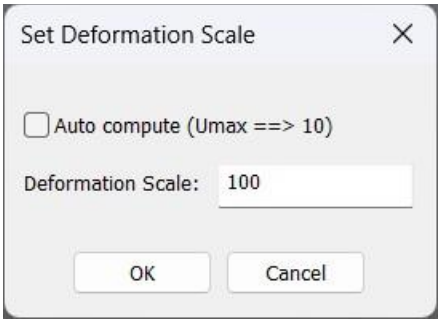


Figure 43: The Set Deformation Scale dialog box

4.6.2 The color mapping of field variables

In most engineering analysis software, variations of field quantities are visualized using color mapping techniques. These techniques rely on a color palette that may be either continuously graded or divided into discrete intervals. Each color range in the palette corresponds to a specific interval of the field variable.

This approach facilitates the graphical representation of physical distributions and enables the user to identify geometric regions with high stress concentrations, which play a crucial role in the design process.

GABES uses a color mapping with a continuously graded rainbow palette shown in Figure 44.

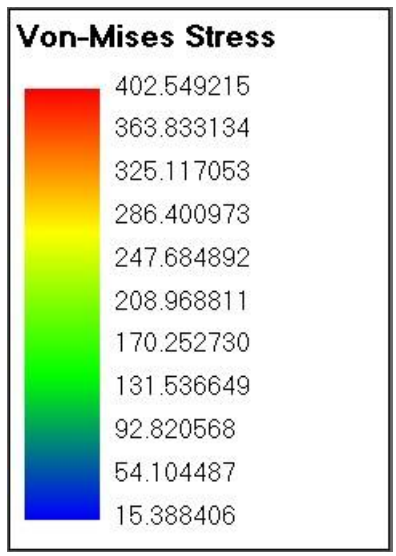
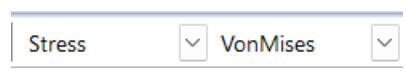


Figure 44: The GABES continuously graded color palette



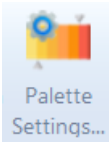
The various field variables that can be color mapped are selected by using the two right most adjacent combo boxes in the **View toolbar**. The left combo box selects the group of variables, whereas the second at the right selects the actual variable to be color mapped. The contents of these two combo boxes are summarized in the Table 14 below.

Table 14: The available field variables for color mapping

Variable group (Left combo box)	**Variable (Right combo box)**	**Variable description**
Nothing	-	No color mapping
Displacement	U1	Directional displacement components
	U2	
	U3	
	Magnitude	The magnitude of the displacement vector

Stress	Sig_11	The stress tensor components
	Sig_22	
	Sig_33	
	Sig_12	
	Sig_23	
	Sig_13	
	VonMises	The Von-Mises equivalent stress
Strain	Eps_11	The strain tensor components
	Eps_22	
	Eps_33	
	Eps_12	
	Eps_23	
	Eps_13	

The numerical formatting of the field variable ranges displayed on the color palette can be modified using the **Palette Settings…** button, which opens the **Palette Settings** dialog box.



As shown in Figure 45, this dialog allows the user to adjust the number formatting of the values displayed on the color palette. Tick values may be shown either in floating-point format or in scientific (exponential) notation, with a user-defined number of decimal digits.

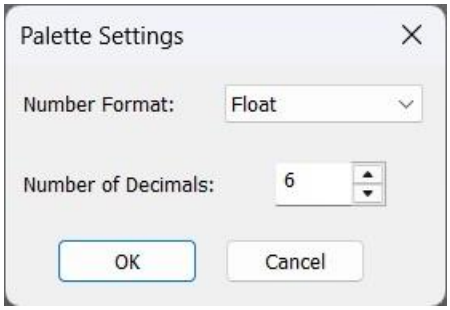
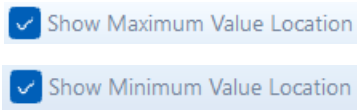


Figure 45: The Palette Settings dialog box

4.6.3 The display of glyphs to locate the max. and min. field values locations

The color mapping of field variables, as shown in the section above, allows the user to visualize the actual distribution and spatial variation of any field variable. Engineers are particularly interested in identifying the maximum and minimum values, such as stress, to verify the validity of their design using appropriate failure criteria. Equally important is determining the geometric locations of these extrema, which provides insight into the model's critical zones. This information can guide local modifications to the geometry, enabling designers to optimize the structure until a safe and reliable model is achieved for the intended operating load range.



In GABES, marking the locations of these critical extrema can be performed using the two check boxes, **Show Minimum Value Location** and **Show Maximum Value Location**, located at the far right of the **BEM Analysis** tab on the ribbon. When **Show Maximum Value Location** is checked, the location of the maximum value is highlighted with a glyph in the form of a bright yellow wireframe sphere, as shown in Figure 46. When **Show Minimum Value Location** is checked, the location of the minimum value is highlighted with a glyph in the form of a bright magenta wireframe sphere, as shown in Figure 47.

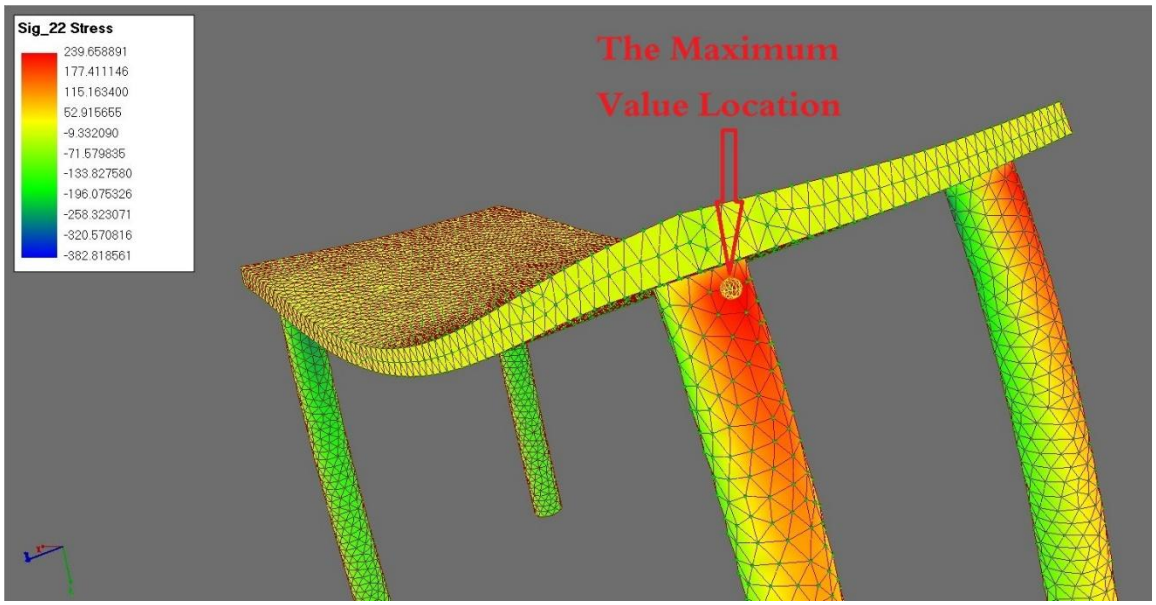


Figure 46: The glyph of the maximum value location

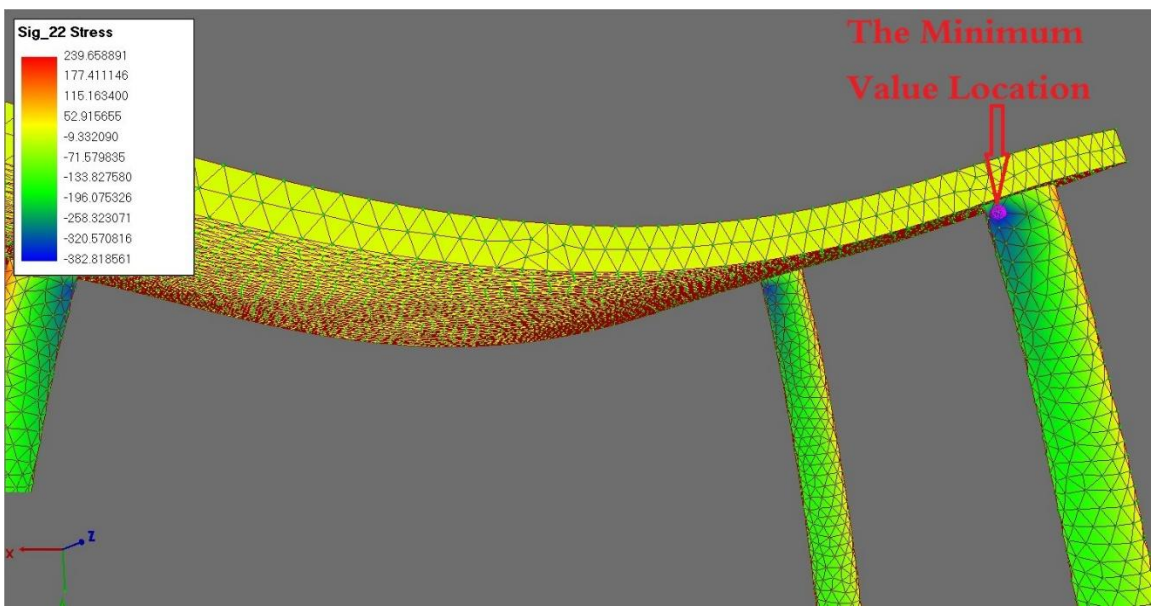
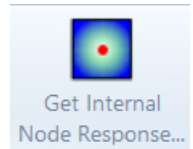


Figure 47: The glyph of the minimum value location

4.6.4 The calculation of the response in a given domain point



After the resolution of the boundary value problem, the user can recover any field variable at any point, either on the boundary or within the interior of the domain. This is accomplished using the **Get Internal Node Response…** button, which opens the **Domain Node Response** dialog box, as shown in Figure 48.

The user can first select the reference frame and the coordinate system (Cartesian, Cylindrical, or Spherical) for the point of interest. Once the reference frame is chosen, the corresponding gizmo is displayed in the 3D scene, providing a visual guide for positioning the track point. The user can then enter the relative coordinates of the point directly. Clicking the **Apply** button displays the point as a red dot on the model, as illustrated in Figure 49.

To improve visibility, the user can zoom in using the mouse wheel, and switch the mesh fill mode to wireframe. This is particularly useful when the point is located inside the domain, allowing the user to clearly see and adjust the location relative to the surrounding geometry.

Once the point is defined, the user can select one or more variables to compute at that location. Clicking the **Calculate** button evaluates the selected variables and displays the results in the corresponding edit boxes. The dialog also allows repeated calculations at different points without closing the window, enabling the user to efficiently probe multiple locations in sequence.

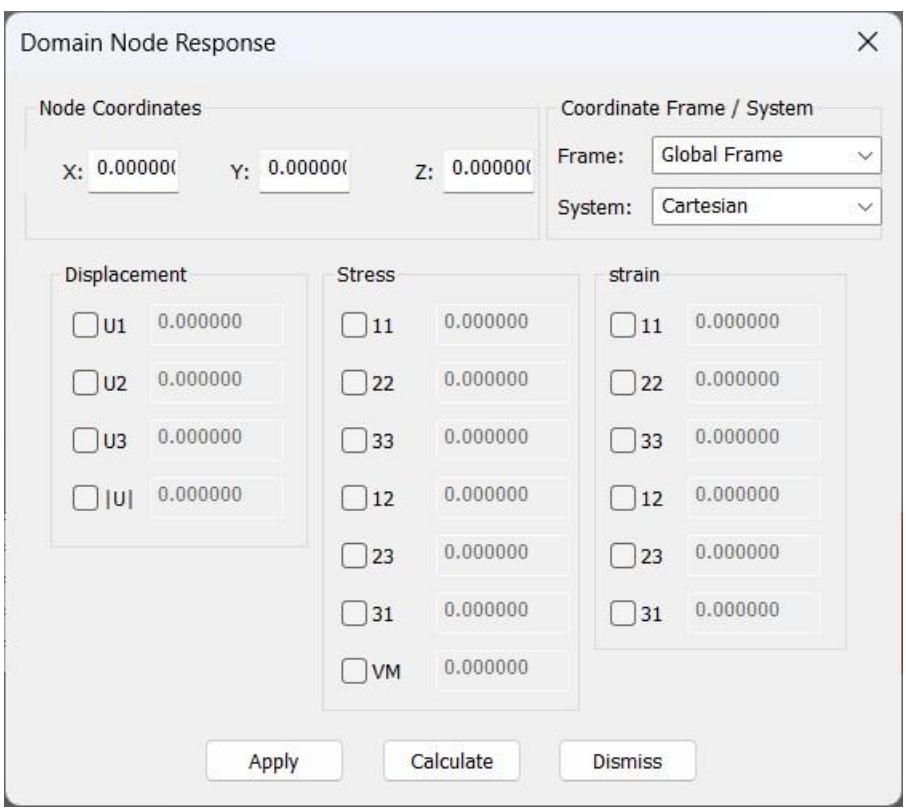


Figure 48: The Domain Node Response dialog box

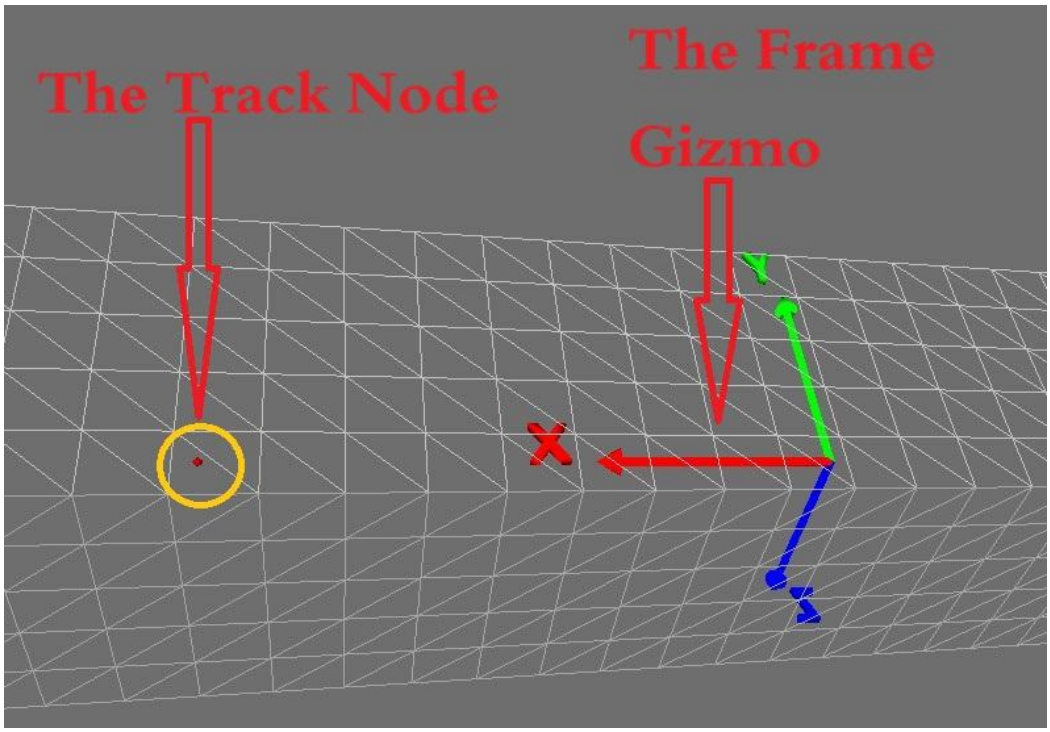
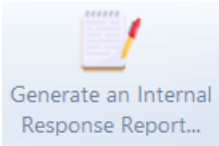


Figure 49: Display of the track node on the 3D model

4.6.5 Compute responses at multiple points and generate tabulated reports



Just like defining a point in the domain and calculating its response, the user can extend this procedure to multiple points to obtain a more comprehensive view of the internal field variables. Using the **Generate an Internal Response Report…** button, several points can be specified, and the corresponding variables are computed at each location. The results are then collected and saved in a tabulated format as a (.csv) file, which can be easily opened and reviewed in spreadsheet software such as Microsoft Excel.

Clicking the **Generate an Internal Response Report…** button opens the **Generate Internal Response Report** dialog box (Figure 50). The user first selects the reference frame and the coordinate system. Once selected, the corresponding gizmo appears in the 3D scene to assist in positioning the track points. The relative coordinates of the points can then be defined using one of two methods (the choice of the method is by activating the correponding radio button):

<u>First method: Using an input file</u>

This method consists of preparing a text file containing the coordinates of all points expressed relative to the selected reference frame. The prepared file must conform to the following format:

- The first line specifies the total number of points.
- Each subsequent line contains the three coordinate components of one point, separated by spaces.

- For Cartesian coordinate system: the components are (x, y, z).
- For Cylindrical coordinates system: the components are (R, θ, z).
- For Spherical coordinates system: the components are (R, θ, φ).
- The angles θ and φ are expressed in degrees, with θ ∈ [−180°, 180°] and φ ∈ [0°, 180°].

Example of an input file:

73 *(The Number of track nodes)*

15 -180 0 (Node 1)

15 -175 0 (Node 2)

15 -170 0 (Node 3)

... .

... .

N.B The text colored in green are just comments, and should not appear in the input file

When the **Use an input file** radio button is selected the input file edit box and the corresponding open file button activates. Clicking the button will open the standard **Open File** dialog box, allowing the user to browse and select the desired input file.

<u>Second method: Generate a grid</u>

Instead of preparing an input text file, the user can generate a structured grid of points interactively. When the **Generate Grid** radio button is selected, the **Open Grid Generation…** button becomes enabled. Clicking this button opens the **Generate Point Grid** dialog box (Figure 51).

This dialog is similar to the **Create Range Subset** dialog described in Section 4.3.2.3. It does not allow selection of the reference frame or coordinate system; instead, it inherits these settings from its parent dialog box (**Generate Internal Response Report**). For each coordinate component, the dialog provides two edit boxes to define the minimum and maximum range values, along with two corresponding slider bars for interactive adjustment. An additional edit box is available for specifying the number of points across the defined range of each direction.

For example, in Cartesian coordinates, the user can define the ranges in the x, y, and z directions and specify the number of points along each axis range. Defining a grid of (5 × 3 × 1) results in a total of 15 points distributed within the specified ranges. The same procedure applies when using Cylindrical or Spherical coordinate systems, where the ranges and the number of points are defined with respect to their corresponding coordinate components, namely (R, θ, z) for cylindrical coordinates and (R, θ, φ) for spherical coordinates.

Figure 52 shows an example of a cylindrical coordinate grid consisting of 255 points arranged as (3 × 17 × 5), i.e., 3 points in the radial direction, 17 points in the angular (circumferential) direction, and 5 points in the axial direction (along the hole axis).

Once the points are defined, the user must select the field variables to be evaluated at these locations. Clicking **OK** prompts, the user to choose the save location for the tabulated (.csv) output file. This file can then be opened in spreadsheet software, such as Microsoft Excel, for data analysis, visualization, and chart generation.

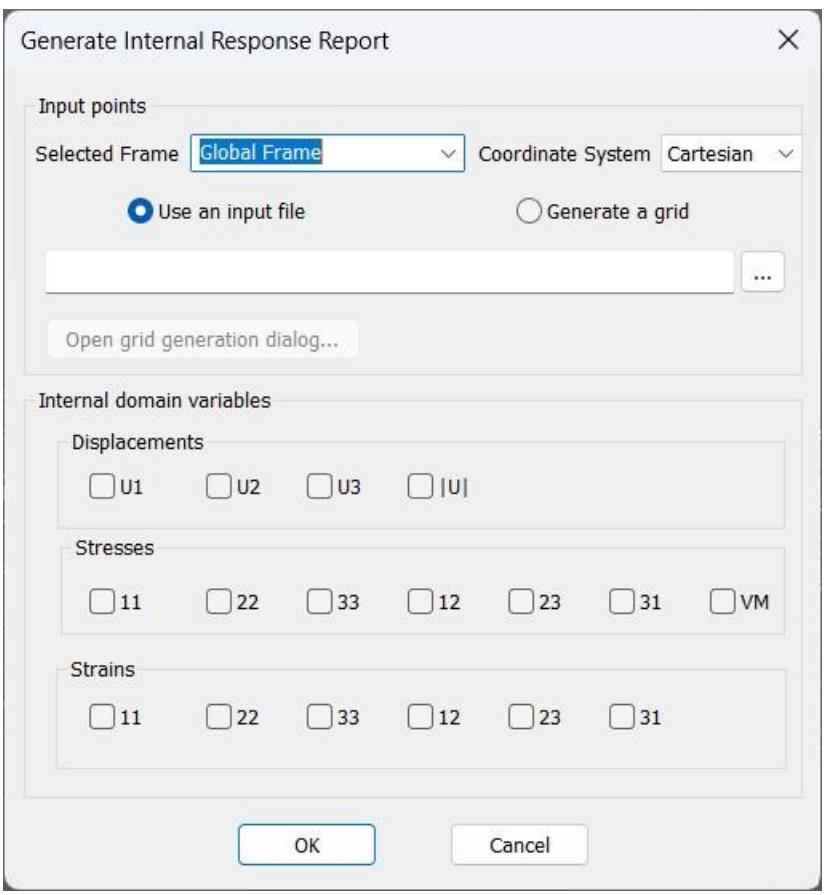


Figure 50: The Generate Internal Response Report dialog box

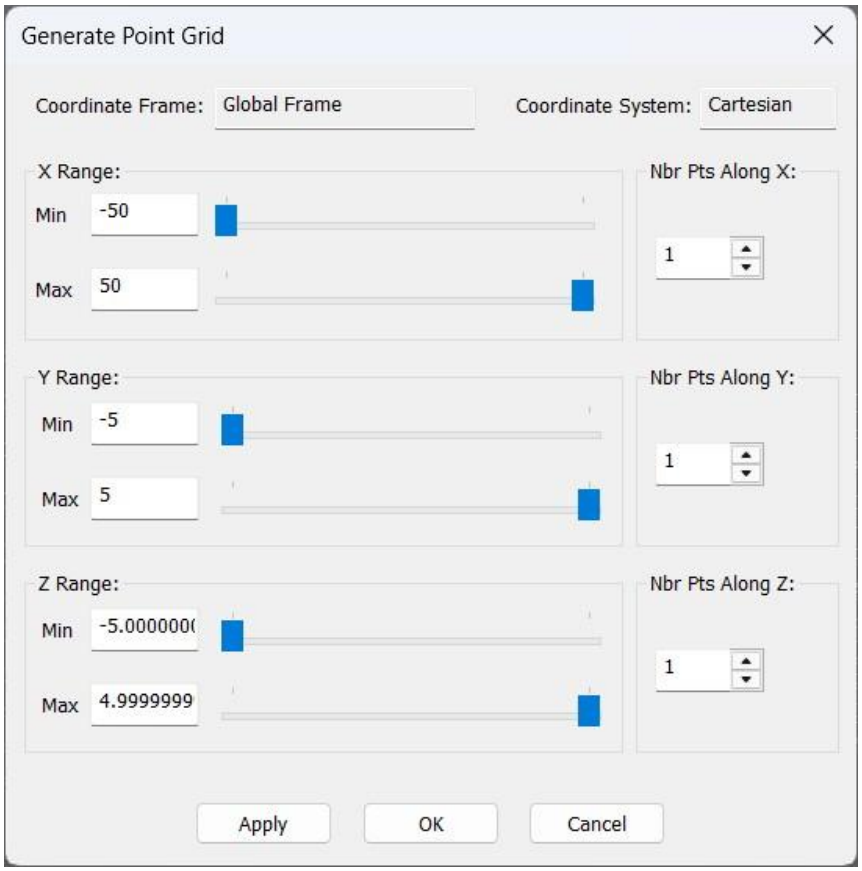


Figure 51: The Generate Point Grid dialog box

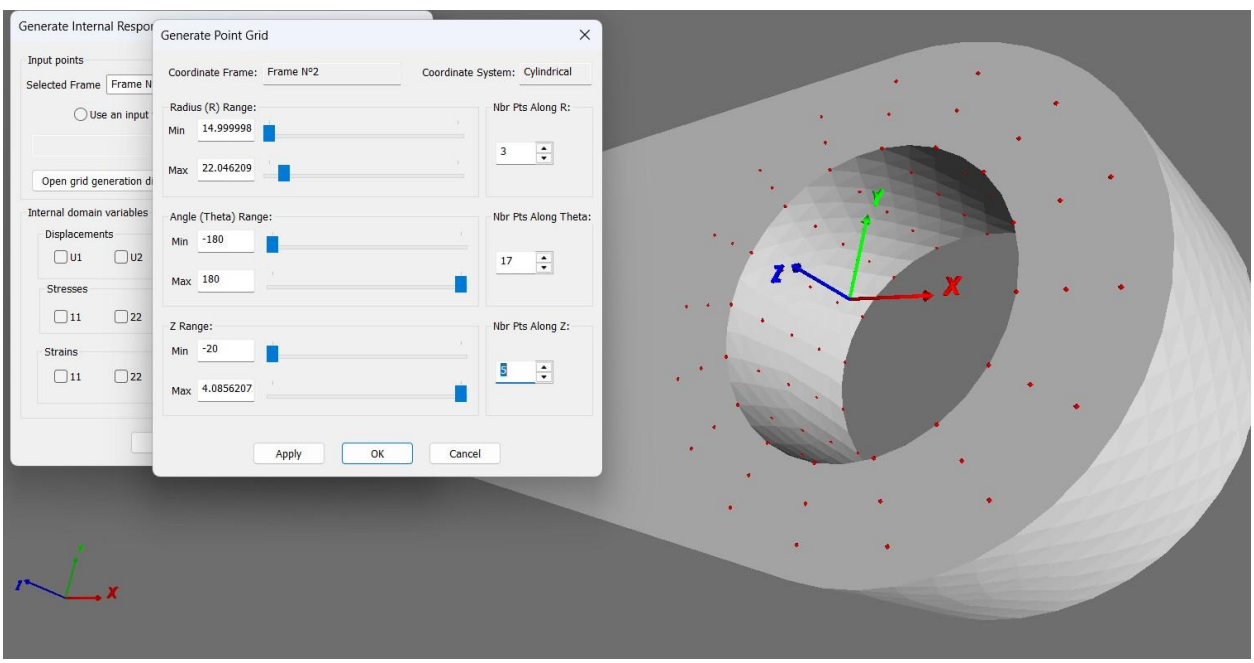


Figure 52: An example of a cylindrical grid